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MACHINE APPLICATION

Vertical Form Fill Seal Machine V1.0

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1. General Introduction

1.1. Overview

With the progress of science and technology, the level of automation in industrial machines continues to increase. In this context, the packaging machine industry is getting more competitive, and the trend is toward more automated, smarter, and more efficient equipment. The vertical packaging machine is a type of packaging equipment mainly used for filling and packaging materials. It can automatically perform the entire packaging process, including bag making, measuring, filling, and sealing.

Vertical packaging machines can be classified by material type, such as liquid or granule, and by measurement method, including weight, flow, and volume. This article covers how the S7-1200 G2 controls a vertical liquid packaging machine that uses screws for volume dosing, including bag making, measuring, filling, and sealing.

This application is based on the typical process and structure of a vertical liquid packaging machine, including unwinding, film pulling, vertical sealing, filling, and horizontal sealing. It includes two parts: the PLC program and the HMI screen.

Figure 1-1: Vertical Form-Fill-Seal machine

1.1.1. Machine Process

The Vertical Form Fill Seal (VFFS) machine consists of the following main components: an unwinder motor, foil feed axis, filling axis, cross sealing axis, vertical sealing jaws, cross sealing jaws, a tension sensor, and a color code sensor.

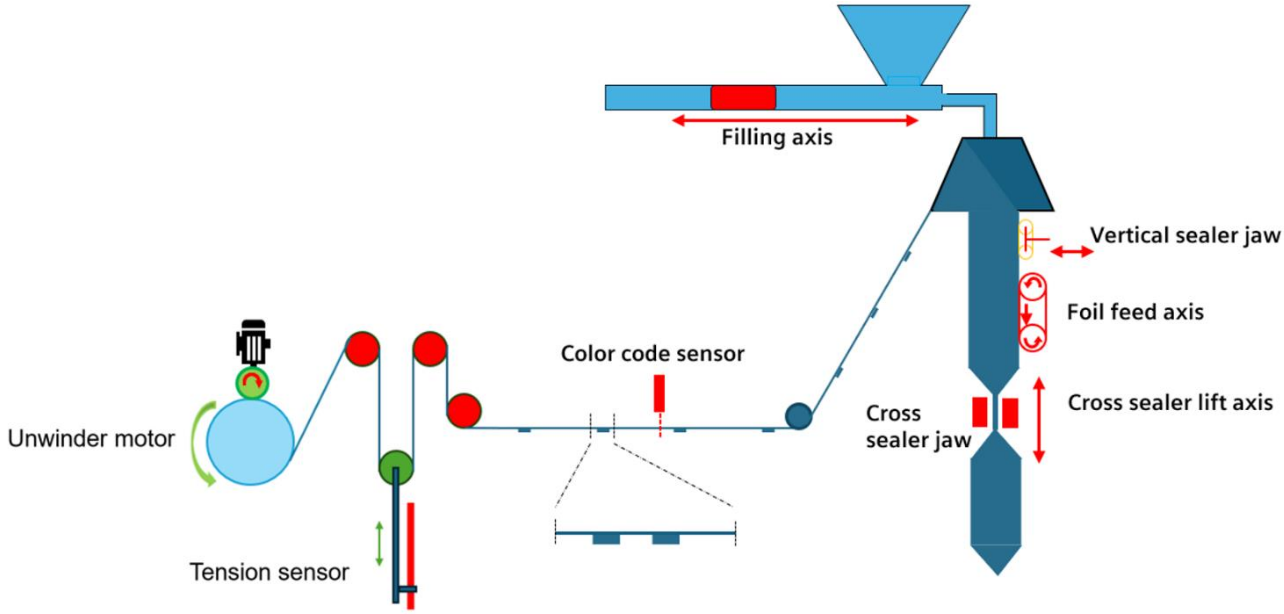


Figure 1-2: Schematic diagram of vertical form fill seal machine

The cross sealing mechanism is the core component of the machine. It combines sealing and cutting, and it moves in sync with the foil to seal and cut at precise intervals and fixed positions.

When the machine is running, the foil feed axis pulls the foil continuously and the vertical sealing cylinder, shapes the foil into a tube. The color code sensor reads the marks on the foil and based on the distance between the marks and the machine calculates the length of each product. Then, the cross sealing mechanism moves to the center of each marked area to seal and cut the foil.

Additionally, the machine has a tension control system that adjusts how the film is unwinding and winding to make sure the packaging is good quality.

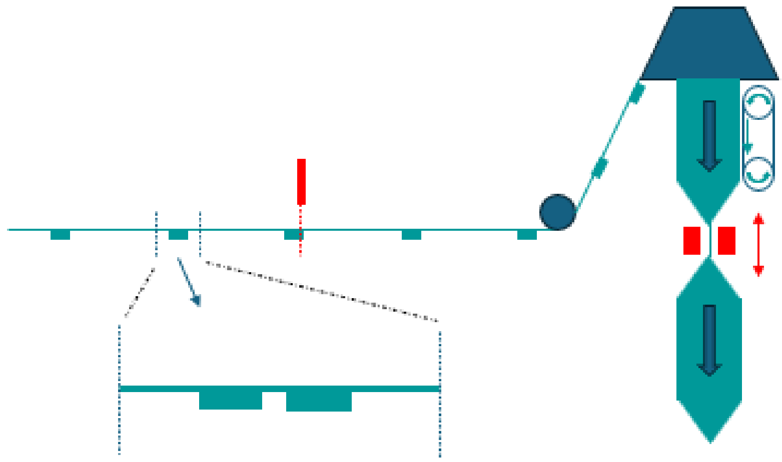


Figure 1-3: Example of color label processing

1.1.2. Introduction to process challenges

During machine operation, the tension sensor continuously sends the actual film tension value via an analog signal. Based on this feedback, the system adjusts the winding and unwinding processes accordingly. In this application, only basic logic control for rewinding and unwinding is implemented. However, if needed, more advanced and precise control functions can be implemented in practice. For more information, please refer to *the SIMATIC Curl and Tension Control Library (LConSMC).*

Additionally, various machine operation factors such as tension and manufacturing deviations can cause discrepancies between the actual position of the color marks and their theoretical values. If these deviations are ignored during production, cumulative errors may occur, potentially resulting in a large number of defective products. To prevent this, it is essential to calculate the actual bag length for each product based on the trigger time of the color mark sensor. If the required accuracy or performance in actual production is not achieved with this approach, refer to the *SIMATIC LPrintMark library (LPrintMark)* for more advanced solutions.

1.2. Components used

The following components are the minimum configuration needed to build the Vertical Form Fill Seal Machine V1.0.

Hardware and software components can be tailored to your specific needs, ensuring good operation of the machine.

Units	Component	Short designation	Article Number	Firmware Version
1	PLC SIMATIC S7-1200 G2	CPU 1214C AC/DC/Rly	6ES7 214-1HH50-0XB0	V 1.0
1	Analog Input Module	SM 1231 AI8	6ES7 231-4HF50-0XB0	V1.0
1	Unified Comfort Panel	MTP1200 Unified Basic	6AV2 123-3MB32-0AW0	Not Applicable
1	Servo Cross Sealer Lift Axis	SINAMICS S200 PN	6SL5510-1BB10-1AF0	V 6.3
1	Servo Filling Axis	SINAMICS S200 PN	6SL5510-1BB10-1AF0	V 6.3
1	Servo Foil Feed Axis	SINAMICS S200 PN	6SL5510-1BB10-1AF0	V 6.3
1	Synchronous motor Cross Sealer Lift Axis	1FL2 Synchronous motor	1FL2102-2AG01-1MC0	Not Applicable
1	Synchronous motor Filling Axis	1FL2 Synchronous motor	1FL2102-2AG01-1SC0	Not Applicable
1	Synchronous motor Foil Feed Axis	1FL2 synchronous motor	1FL2102-2AG01-1MC0	Not Applicable
3	Motion connection cables	MOTION-CONNECT 350 Power cable	6FX3502-7CD01-1AF0	Not Applicable
1	Analog sensor for tension control	Generic analog sensor	Not Applicable	Not Applicable
1	Digital color code sensor	Generic digital sensor	Not Applicable	Not Applicable

Table 1-1: Hardware components

Component	License	Version specifications	Article Number Download
SIMATIC STEP 7 V20 Engineering Software	SIMATIC STEP 7 Basic V20 Floating License	V20 Update 3	6ES7822-0AE24-0YA5
SIMATIC WinCC Unified Engineering V20	WinCC Unified Basic ES V20	V20 Update 2	6AV2151-0XA02-4LA5
SINAMICS Startdrive V20 Engineering Software	Startdrive Basic V20	V20	6SL3072-4LA02-0XG0

Table 1-2: Software Components

The libraries listed below are already included by default in the Global Library **Automation Framework V2.0** https://support.industry.siemens.com/cs/ww/en/view/109817223. You can customize the selection, adding or removing libraries as needed.

Component	Version	SIOS Link
Axis Control (LAxisCtrl)	V1.4.0	https://support.industry.siemens.com/cs/ww/en/view/109925893
Library of Basic Controls (LBC)	V3.1.0	https://support.industry.siemens.com/cs/ww/en/view/109792175
Libraries of PLC data types (LPD)	1.0.0	https://support.industry.siemens.com/cs/ww/en/view/109482396
Library of general functions (LGF)	V5.1.1	https://support.industry.siemens.com/cs/ww/en/view/109479728
SIMATIC OMAC PackML V2022 Mode&State Management and Machine Data Interface (LPMLV2022)	V1.0.0	https://support.industry.siemens.com/cs/ww/en/view/109821198
Lsicar	V5.1	https://support.industry.siemens.com/cs/ww/en/view/109804254
Library for SNTP Server Functionality (LSNTP)	V4.0.0	https://support.industry.siemens.com/cs/ww/en/view/82203451
Library of Unit Control – LUC	V2.0.0	https://support.industry.siemens.com/cs/ww/en/view/109974940

Table 1-3: Library Types

NOTE You can purchase these components from the Siemens Industry Mall.

This machine application consists of the following components:

Component	File name
TIA Portal Project	109989389_VerticalFormFillSealMachine_PROJ_V1_0.zip
Manual	109989389_VerticalFormFillSealMachine_DOC_V1_0_en.pdf
Positioning and Connection	109989389_PositioningAndConnection_en.pdf
Bill of Materials	109989389_BOM.pdf
TIA Selection Tool File	109989389_VerticalFormFillSealMachine_TST_V1_0.zip

Table 1-4:File components

1.3. Description of the project

The Vertical Form Fill Seal Machine provides a TIA Portal project that includes a Simatic S7-1200 G2 PLC, a MTP1200 Unified Basic Panel, and three SINAMICS S200 PN to control the axes.

Device name	Device type	IP Address
VFFSPLC	SIMATIC CPU 1214C	192.168.0.1
VFFSHMI	MTP 1200 Unified Basic Panel	192.168.0.2
CrossSealerLiftAxis	SINAMICS S200PN	192.168.0.3
FillingAxis	SINAMICS S200PN	192.168.0.4
FoilFeedAxis	SINAMICS S200PN	192.168.0.5

Table 1-5: Machine application devices

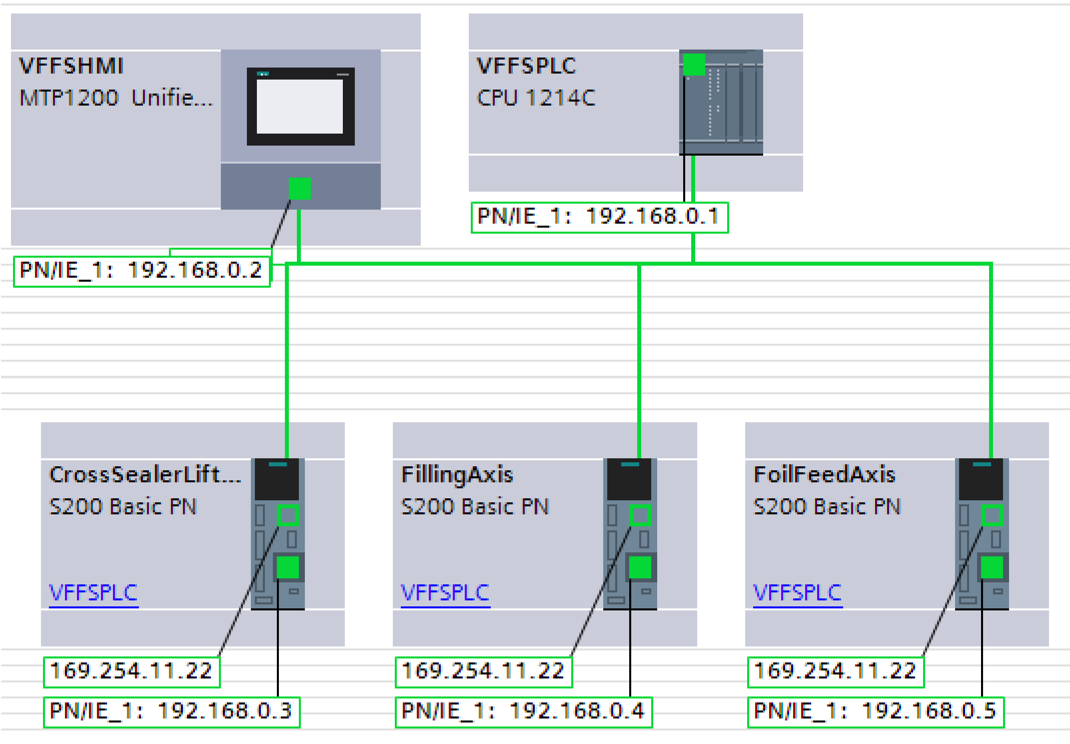


Figure 1-4:TIA Portal Network View

The credentials configured in the project are detailed below:

User	Password	Note
Jones	Siemens!123	Maintenance
Smith	Siemens!123	Administrator

Table 1-6: Credentials

2. PLC Program Structure

The Vertical Form Fill Seal Machine is developed using the Automation Framework Basic (AF_{Basic}).

The Automation Framwork Basic is a framework that provides the basic structure of a PLC and HMI program and a standardized interface for controlling and monitoring the machine status, following the ISA-88 standard. The equipment is divided into three layers from top to bottom according to control: Unit, Equipment Module (EM), and Control Module (CM)

Unit

The unit represents a physical piece of equipment in production, in this case, a Vertical Form Fill Seal Machine. The unit contains equipment modules (EMs) which contain control modules (CMs).

Equipment module:

Once the unit boundary is defined, the lower-level modules of the Equipment Module can be identified. An EM is a functional group of Control Modules that can carry out a limited number of specific processing activities such as dosing, charging or temperature control. An EM is typically centered around a piece of process equipment such as vessel outlet/inlet manifold, process heater etc. The separation of a unit into functional EM should reflect the process actions from the process and procedural control model.

Control module

The CMs are the lowest level grouping of equipment in the physical model that can carry out basic control and have a direct relationship to the type of instrument/device that it is being monitored and/or controlled. The CMs are typically directly connected to IO signals.

This structure allows a flexible combination and configuration of device functions to meet the production needs and application scenarios.

As shown in Figure 2-1, the Vertical Form Fill Seal Machine is divided into one unit, four EMs, and eight CMs.

Each EM includes the following features:

- EM0 is a general part, which is used for the control of the general parts such as the power supply and distribution circuit of the whole equipment.
- EM1 is the feeding and sealing system of the machine, which is divided into 4 CMs: Cross sealing axis, foil feeding axis, cross sealing jaw and vertical sealing jaw. This EM controls all systems to ensure optimal sealing of the packages.
- EM2 is the filling system of the machine, which is divided into 1 CM the filling axis. This EM ensures a continuous feed of product to the machine.
- EM3 is the unwinder system of the machine, which is divided into 2 CMs: Unwinder motor and analog tension sensor. This EM ensures that the film moves with the correct tension to prevent tearing or low quality packaging.

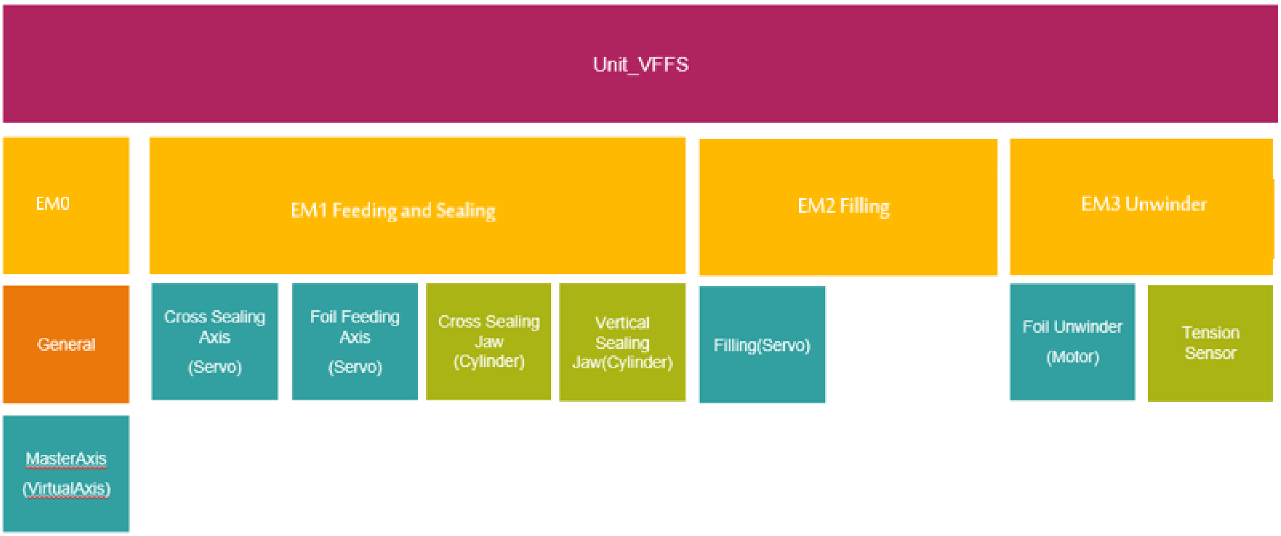


Figure 2-1: ISA-88 equipment division of the Vertical Form Fill Seal Machine

The development of the entire project program follows the OMAC PackML state machine:

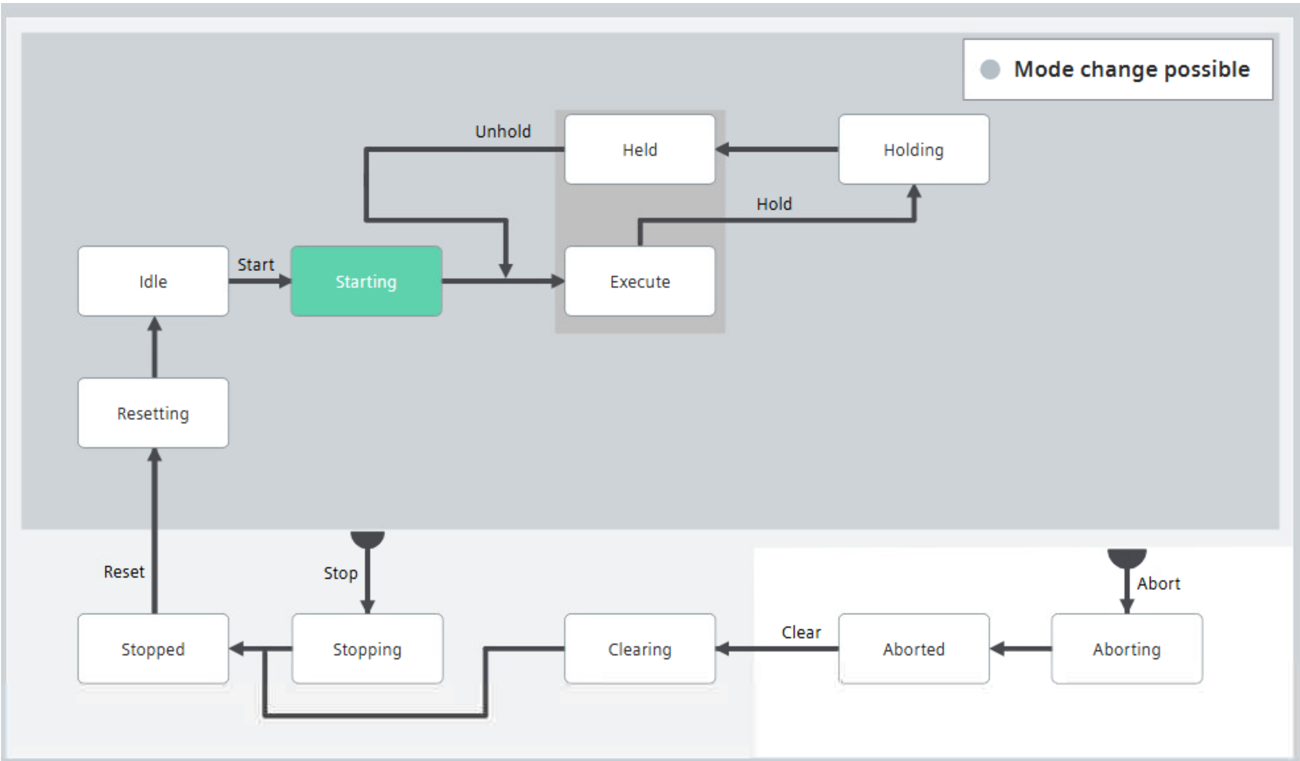


Figure 2-2: Base state model

The following Modes, States and Commands are supported by the OMAC PackML state machine used in the AF$_{Basic}$:

Modes

0	Invalid
1	Production
2	Maintenance
3	Manual
4 .. 15	User definable

States

0	Undefined
1	Clearing
2	Stopped
3	Starting
4	Idle
5	Suspended
6	Execute
7	Stopping
8	Aborting
9	Aborted
10	Holding
11	Held
12	Unholding
13	Suspending
14	Unsuspending
15	Resetting
16	Completing
17	Completed

Commands

0	Undefined
1	Reset
2	Start
3	Stop
4	Hold
5	Unhold
6	Suspend
7	Unsuspend
8	Abort
9	Clear
10	Complete
11	StateComplete

Figure 2-3: Modes, States and Commands that can be used in the AF$_{Basic}$

This state machine can be flexible reconfigured, however, that would imply major changes in the current AF$_{Basic}$ implementation and shall be avoided if possible.

In the Vertical Form Fill Seal Machine, only three operating modes have been configured:

- Production (mandatory): This is the production routine mode. If this mode is selected, the machine executes the relevant logic in response to commands. This commands mainly coming from external systems or entered directly by the operator.
- Maintenance (optional): This mode allows authorized personnel to run the machine independently from other systems. This mode would typically be used for troubleshooting, machine trials or testing operational improvements.
- Manual (optional): This mode allows the direct control of single machine modules. It may be used for commissioning of individual components, fault diagnostics of unplanned technical intervention, etc.

In the same way, not all states and commands have been implemented as they were not required for the defined functionality. To automatically enable or disable the different modes, states, and commands,just set the corresponding bit to "True" or "False" within the "U1_Config [DB195]" data block in the TIA Portal project.

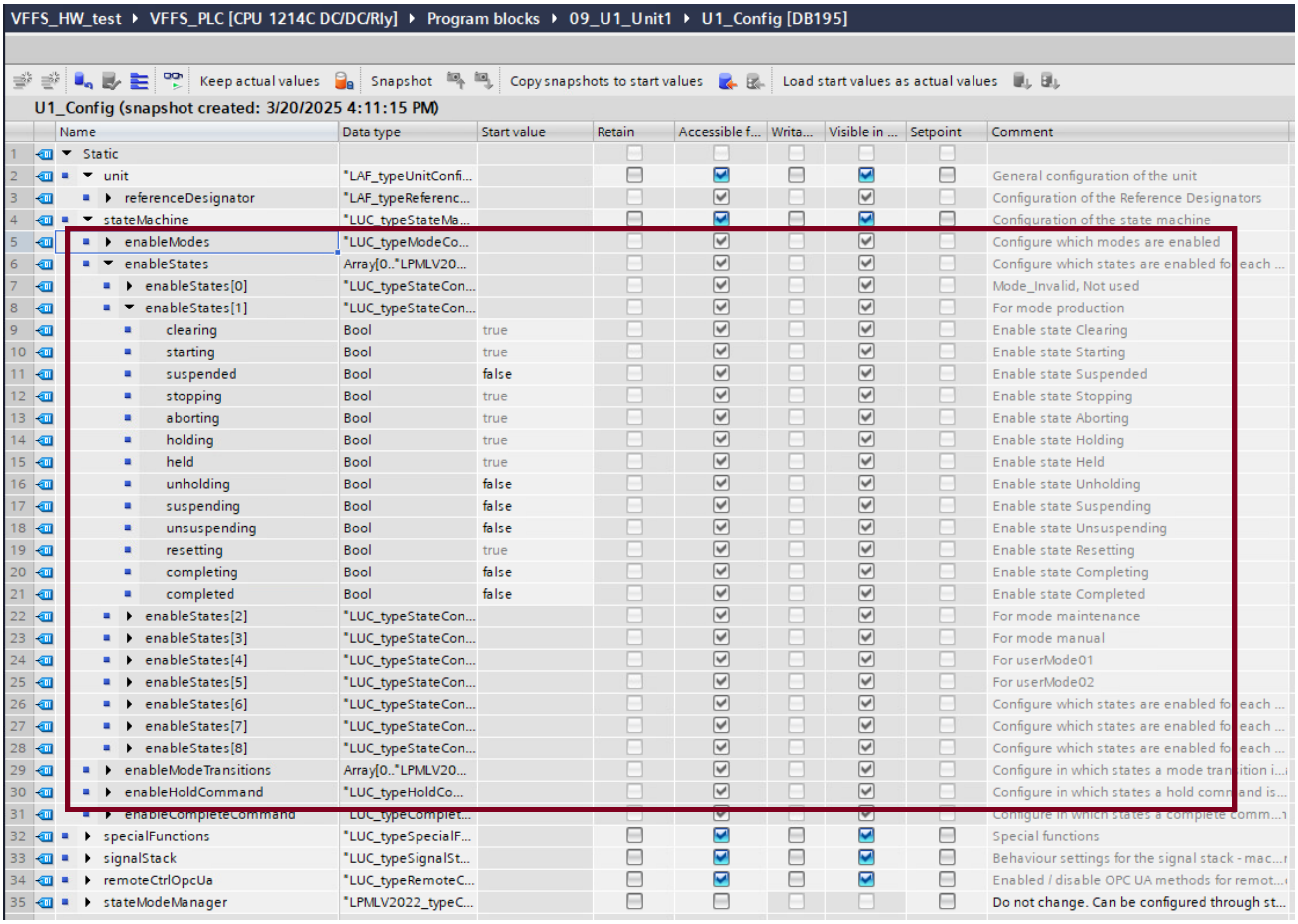


Figure 2-4: State Model Configuration

NOTE	If additional information is required about Automation Framework, please contact Siemens support or consult directly the Automation Framework official documentation: https://support.industry.siemens.com/cs/ww/en/view/109817223

2.1. PLC Program Overview

The PLC contains many of the main functions. Some of the most important nodes are highlighted below:

- Program blocks: Contain all the instructions and logic required to control the machine.
- Technology objects: Preconfigured and reusable blocks used to set up and control advanced functions like Motion Control, axis management or PID regulation.
- Input and output: Devices used are defined within the PLC tags.
- PLC data types: All user-defined data types used in the program.

The above are the main functions, and the rest of the nodes are for general use, they can be implemented according to the needs.

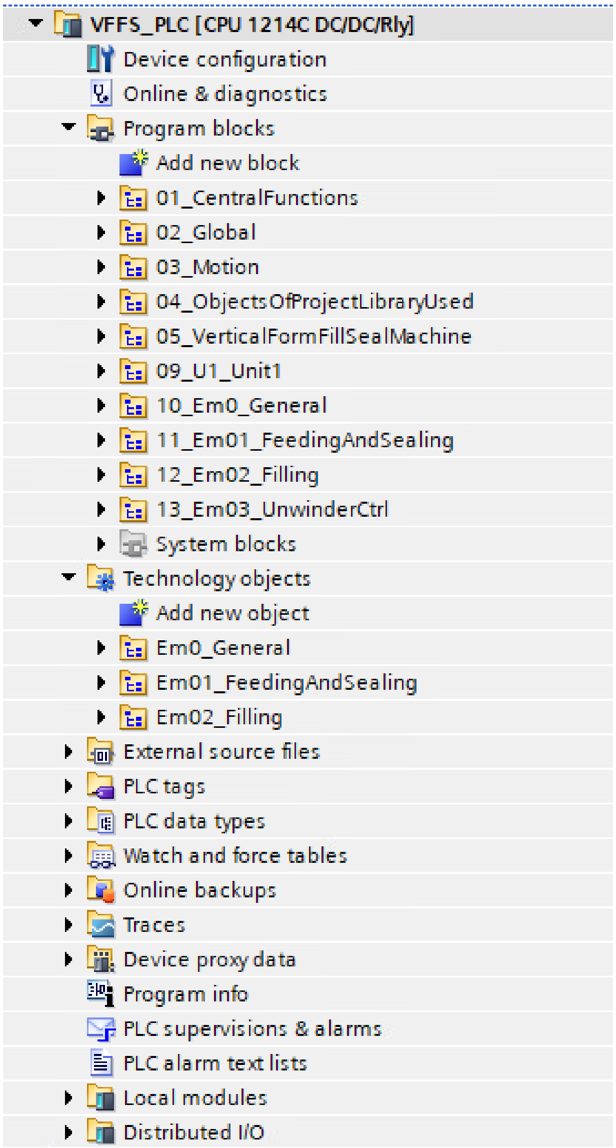


Figure 2-5: Program Overview

The structure of the main program blocks within each folder configured in the PLC is described in detail in the following sections.

NOTE	If additional information is required about Automation Framework, please contact Siemens support or consult directly the Automation Framework official documentation: https://support.industry.siemens.com/cs/ww/en/view/109817223

2.1.1. 01_CentralFunctions

"01_Central Functions" contain system-wide functionality which affects the entire system or is hierarchically above the unit level. They include features, settings, and diagnostics that influence the system globally or across multiple units.

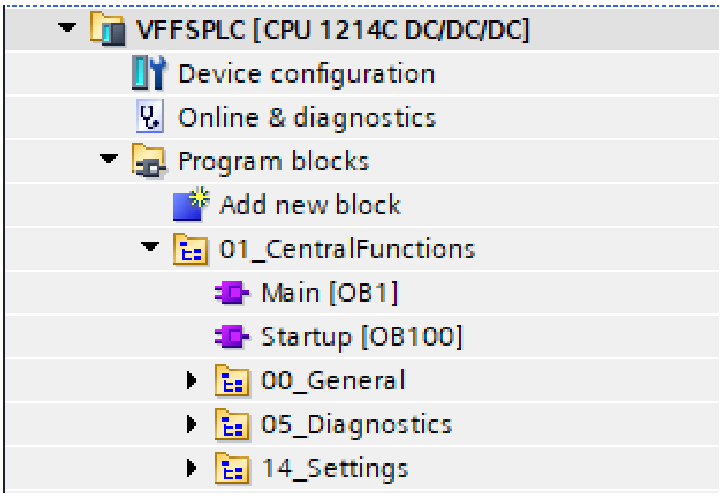


Figure 2-6: Central Functions Overview

2.1.2. 02_Global

The "02_Global" folder contains global data of all units and contains no logic, it is purely passive. It has a relation to ObjectsOfProjectLibraryUsed as it uses PLC data types from the libraries.

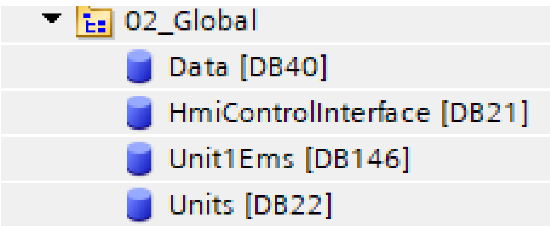


Figure 2-7: Global Overview

2.1.3. 03_Motion

The "03_Motion" folder contains the Motion OBs. These execute the motion Technology Objects like axes or cam outputs. Each Motion OB can only exist once in the entire PLC.

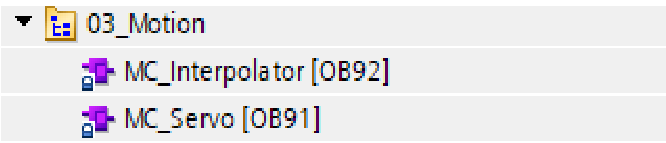


Figure 2-8: Motion Overview

2.1.4. 04_ObjectsOfProjectLibraryUsed

The "04_ObjectsOfProjectLibraryUsed" folder contains the local instances of the libraries that are used in the PLC program. It only contains blocks that are used by others.

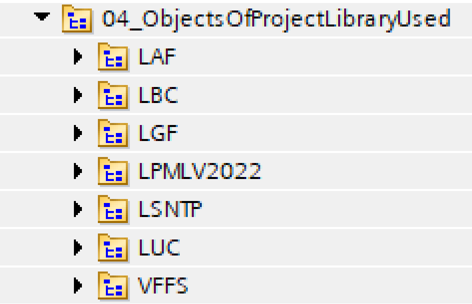


Figure 2-9: ObjectsOfProjectLibraryUsed Overview

2.1.5. 05_VerticalFormFillSealMachine

The "05_VerticalFormFillSealMachine" folder contains the FB's blocks which are custom created by for the process.

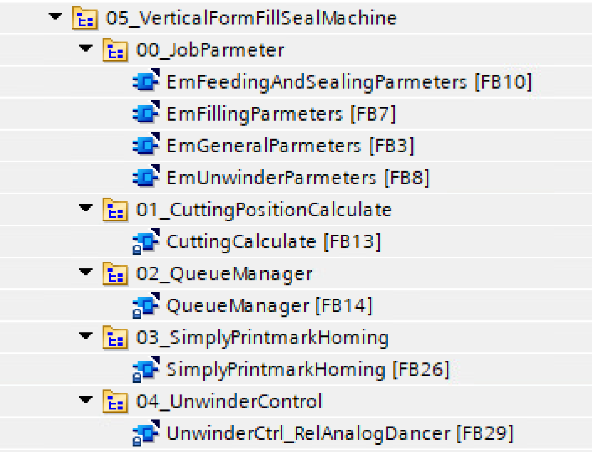


Figure 2-10: Vertical Form Fill Seal Machine Overview

2.1.6. 09_U1_Unit1

The "09_U1_Unit1" describes the general structure, interfaces, commands, and data flow. Primarily, the unit manages the configuration of the mode state manager, handles supplementary signals and processes and manages internal alarms. This Unit contains the following blocks:

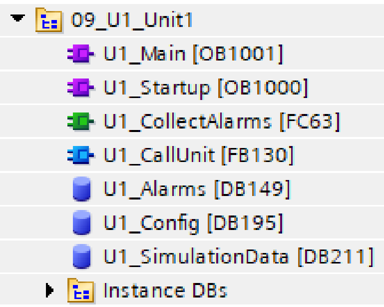


Figure 2-11: Unit1 Overview

2.1.7. 10_Em0_General

The '10_Em0_General' usually contains non-complex logic and no sequences. It includes all common equipment condition monitoring and logic control. In this application, it also contains the virtual master axis control logic.

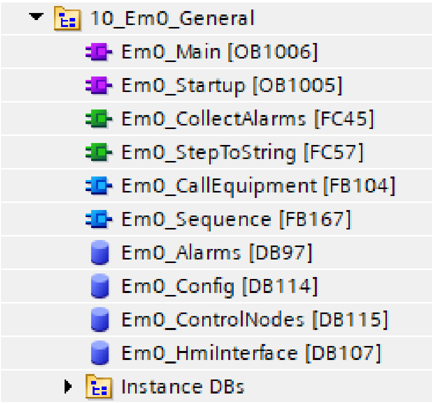


Figure 2-12: Em0_General Overview

2.1.8. 11_Em01_FeedingAndSealing

The "11_Em01_FeedingAndSealing" includes the control of the sealing system. The simulation of the color code sensor and the air cylinder signals for the sealing jaws are also included.

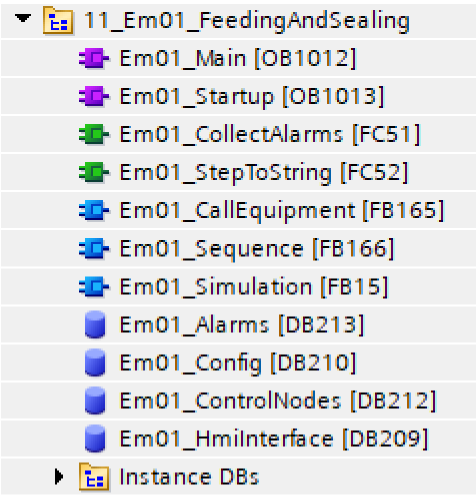


Figure 2-13: Em01_FeedingAndSealing overview

2.1.9. 12_Em02_Filling

The "12_Em02_Filling" contains the filling axis control logic, condition monitoring, and alarm collection.

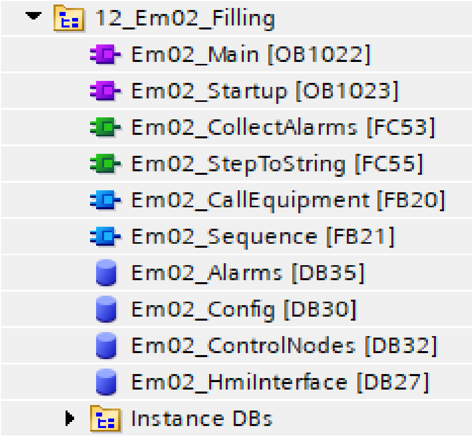


Figure 2-14: Em02_Filling overview

2.1.10. 13_Em03_UnwinderCtrl

The "13_Em03_UnwinderCtrl" contains control for the unwinding motor and tension sensor, including condition monitoring and alarm handling.

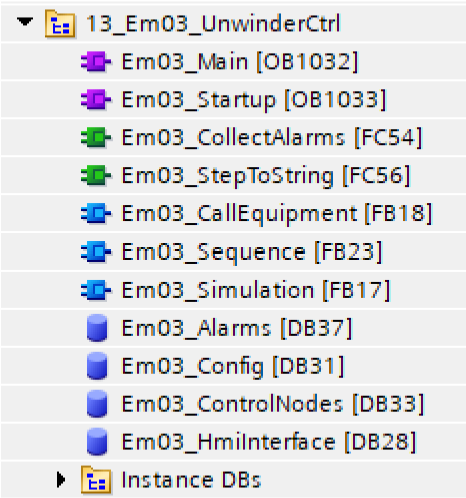


Figure 2-15: Em03_UnwinderCtrl overview

2.1.11. Technology Objects

The Technology Objects control the movement of real objects (e.g., a drive) and notify status information (e.g., the current position). The configuration of the Technology Objects represents the characteristics of the real objects. The configuration data is stored in a data block. Thanks to Technology Objects, the configuration and programming of complex systems can be simplified, improving development efficiency and system reliability.

In this machine, three types of Technology Objects are used:

- TO_PositioningAxis: It is used to position a drive with position control. With the Motion Control instructions, positioning commands are sent to the axis from the user program. In this machine application, this TO controls the virtual master axis. This virtual axis can serve as a leading axis to synchronize multiple real following axes, adding flexibility to the system design.

- TO_SychronousAxis: It contains all the functions of the Positioning Axis technology object. Additionally, the axis can be connected to a master value, allowing the axis to follow the position variation of a master axis in synchronization. In this machine application, the TO_SychronousAxis signals are used to control the feeding, sealing, and foil forming axes.

- TO_Cam: It defines a function f(x) using interpolation points or segments. The cam disc represents the mathematical synchronous relationship between a master and a slave axis. In this machine application, the "crossSealerLiftCam" defines the relationship between the cross sealing axis and the virtual master axis, and the "foilFeedCam" defines the relationship between the foil feeding axis and the virtual master axis.

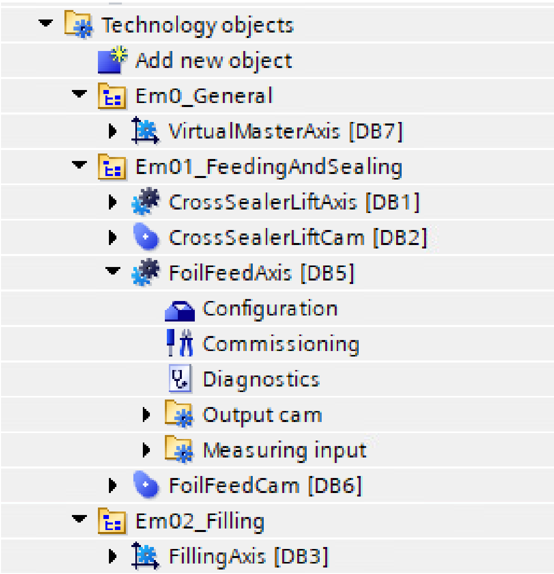


Figure 2-16: Technology Objects Overview

Additionally, this folder also contains the necessary information and configuration for the color code sensor. Because as mentioned in this machine application, the sensor is directly connected to the "Foil Feed Axis" drive rather than to the PLC.

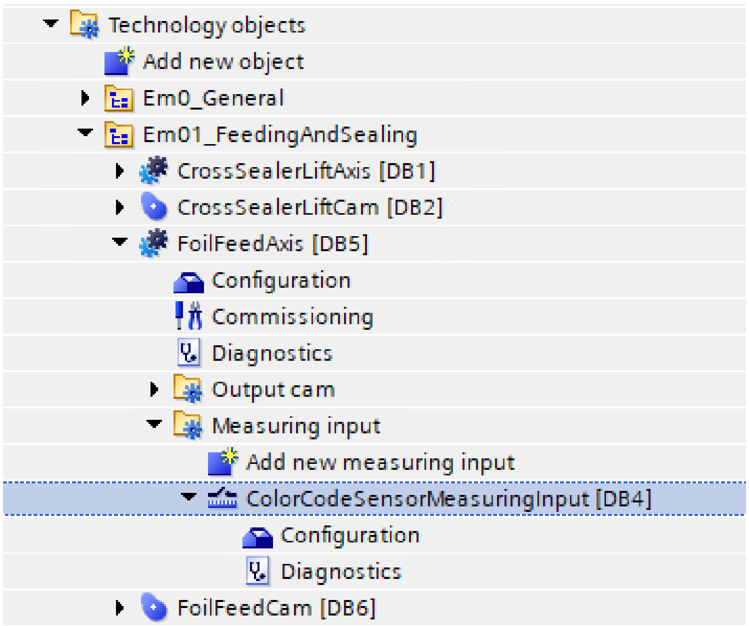


Figure 2-17: Color Code Sensor 1

2.2. Sequence Introduction

As mentioned in previous sections, the program is divided into one Unit that contains four EMs according to ISA-88: The EM0, EM1, EM2 and EM3.

The process performed within each EM is illustrated in the Figure 2-18.

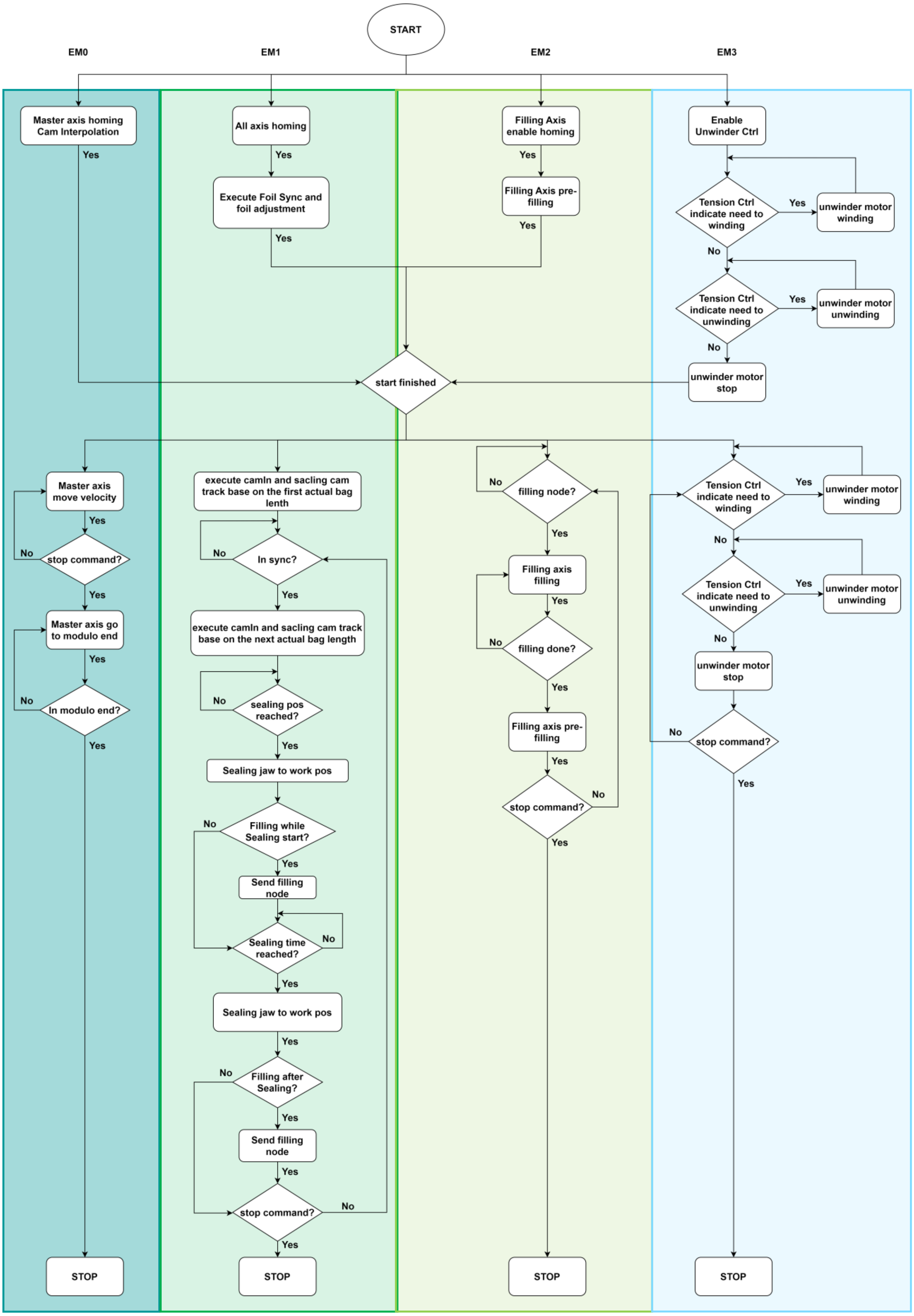


Figure 2-18：Sequences

2.3. Vertical Form Fill Seal Machine Program Blocks

In this machine application, four custom function blocks have been created to control the process:

- CuttingCalculateCn block: It is used to calculate the actual bag length based on the results from the "MC_MeasuringInputCyclic" instructions of the color code sensor connected to the Foil Feed Drive.
- SimplyPrintmarkHomingCn: It is used to adjust the foil feed axis and establish the positional relationship between the foil feed axis and the actual form position.
- QueueManager: Is used to save and extract actual bag length form a queue.
- UnwinderCtrl: It controls the unwinder motor to wind or unwind the film based on the tension feedback from the tension sensor, keeping the film tension within a specified range.

2.3.1. CuttingCalculateCn

This block is used to calculate the actual bag length based on the parameters "Rising count", "Rising position", "Falling count", and "Falling position". These parameters are obtained through the instruction "MC_MeasuringInputCyclic".

When the rising edge count increases by one, the function stores the corresponding rising position in a buffer. Then, when the falling edge count increases by one, a print mark area is detected. Once the number of detected print mark areas reaches the value specified in printMarkCount, a color code area is recognized.

The actual bag length is calculated as the distance between the midpoints of two consecutive color code areas.

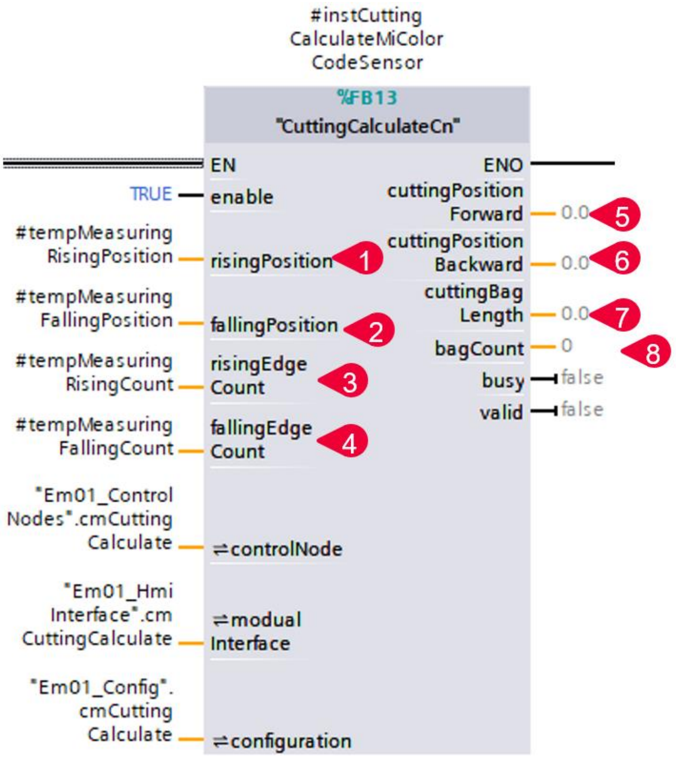


Figure 2-19:CuttingCalculateCn Overview

Number	Signal	Main Functionality	Description
1	risingPosition	Rising Position	The rising edge position from the instruction
2	fallingPosition	Falling Position	The falling position from the instruction
3	risingEdgeCount	Rising Edge Count	The rising edge count from the instruction
4	fallingEdgeCount	Falling Edge Count	The falling edge count from the instruction
5	cuttingPositionForward	The cutting position of the front of the current bag	The position where the film should be cut at the front
6	cuttingPositionBackward	The cutting position of the end of the current bag	The position where the film should be cut at the back
7	cuttingBagLength	The length of the current bag	The actual bag length calculated

Table 2-1: Function Block Description

2.3.2. SimplyPrintmarkHomingCn

This block detects the measurement input signal from the color code sensor and generates a jog control signal. When the sensor receives the signal, it immediately stops the jog command, allowing the system to determine the positional relationship between the film and the axis.

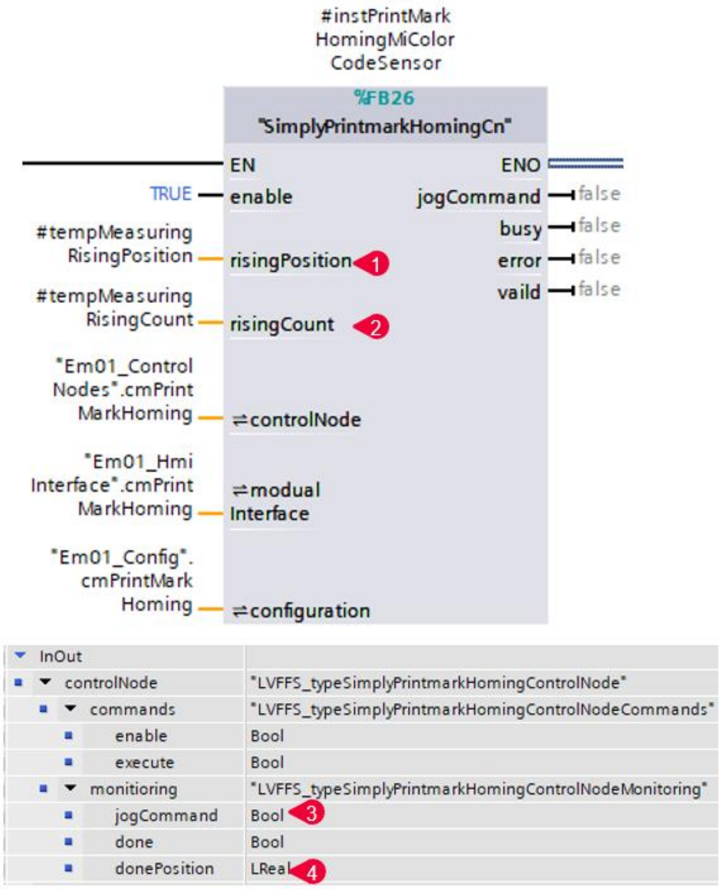


Figure 2-20: SimplyPrintmarkHomingCn Overview

Number	Signal	Main Functionality	Description
1	risingPosition	Rising Position	The rising edge position from the instruction
2	risingCount	Rising Edge Count	The rising edge count from the instruction
3	controlNode.monitoring. jogCommand	Outputs jog commands	When this signal was TRUE, specified that the foil need to keep the film move to sealing position
4	controlNode.monitoring. donePosition	Homing position	This position specified the position of foil feed axis relative to original point of foil

Table 2-2: Function Block Description

2.3.3. QueueManager

This block is used to save and extract the actual bag length form a queue.

The "enqueueItem" will be stored into queue if the "enqueueCommand" was triggered.

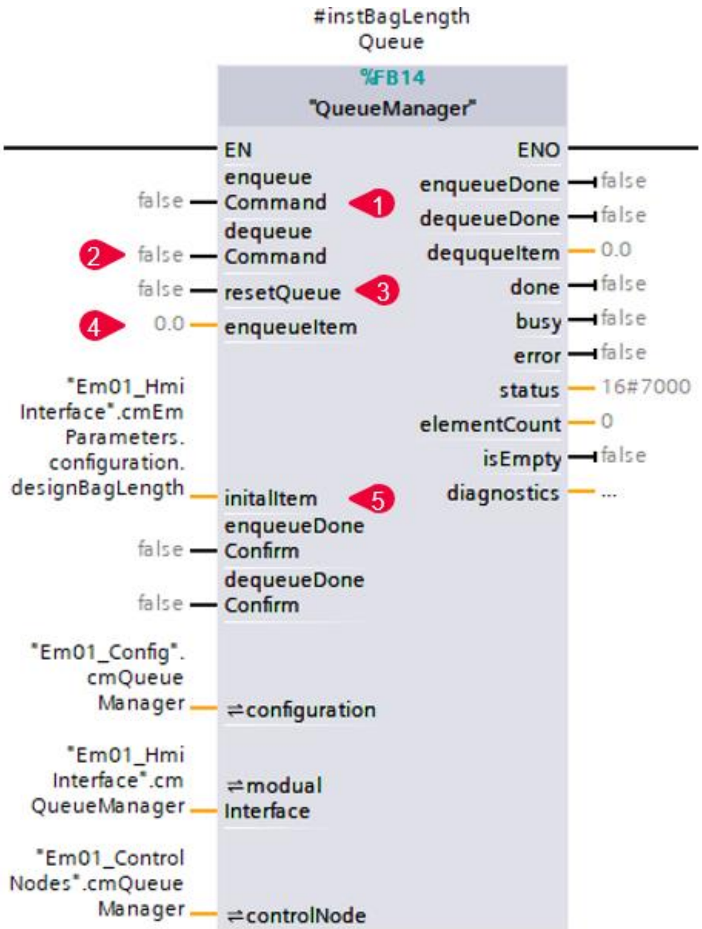


Figure 2-21: QueueManager Overview

Number	Signal	Main Functionality	Description
1	enqueueCommand	Command to enqueue	The rising edge triggers the insertion of a new element to the end of the queue
2	dequeueCommand	Command to dequeue	The rising edge triggers the removal of an element
3	resetCommand	Reset command to reset the queue	The rising edge triggers a reset of the current queue
4	enqueueItem	The item that needed to be enqueue	The element that was need to enqueue
5	initialItem	The initial item in queue after reset	The initial element of the queue

Table 2-3: Function Block Description

2.3.4. UnwinderCtrl

This block controls the unwinding motor using feedback from the tension sensor, which sends an analog signal to indicate the tension level. If the tension is too high, the motor unwinds the film. If the tension is too low, it winds it back.

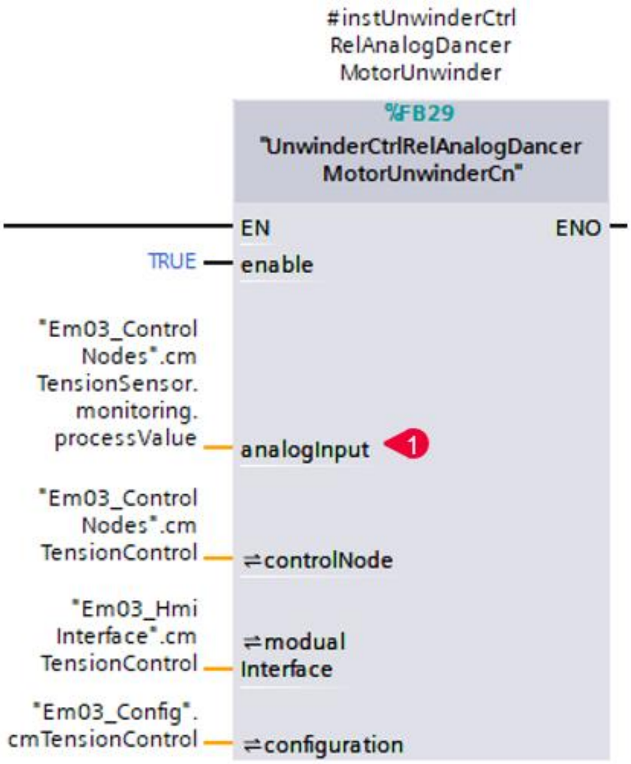


Figure 2-22: UnwinderCtrl_Analog Input Overview

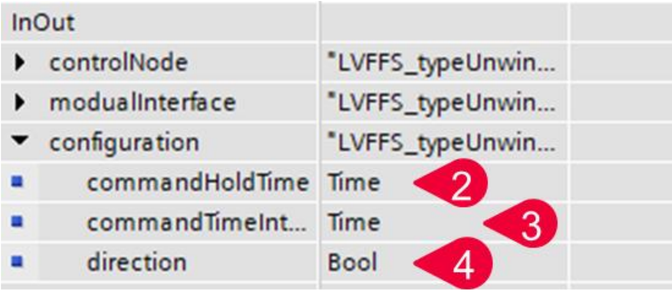


Figure 2-23: Screen shot

Number	Signal	Main Functionality	Description
1	analogInput	Input value, indicated the tension level	Indicates the analog value of the tension sensor in real numbers
2	configuration.commandHoldTime	Unwinding/winding command holding time	Refers to the time that the control command keeps the output after the overly loose or overtight signal from the tension pendulum feedback disappears. This applies only when the tension pendulum is digital.
3	configuration.commandTimeInterval	Unwinding/winding command interval time	The minimum interval between forward and reverse command switching
4	configuration.direction	Unwinding direction	Controls the direction of the command. If TRUE, the analog input increases, indicating that the tension level is decreasing.

Table 2-4: Function Block Description 1

3. HMI Introduction

3.1. Overview of the main Screen

The HMI project is also configured based on Automation Framework Basic. It provides a modular and customizable HMI template for WinCC Unified to reduce programming efforts.

The following chapter will introduce the HMI details.

NOTE For more details, see the Automation Framework documentation: https://support.industry.siemens.com/cs/ww/en/view/109817223

or the LBC (Library of Basic Controls) documentation: https://support.industry.siemens.com/cs/ww/en/view/109792175

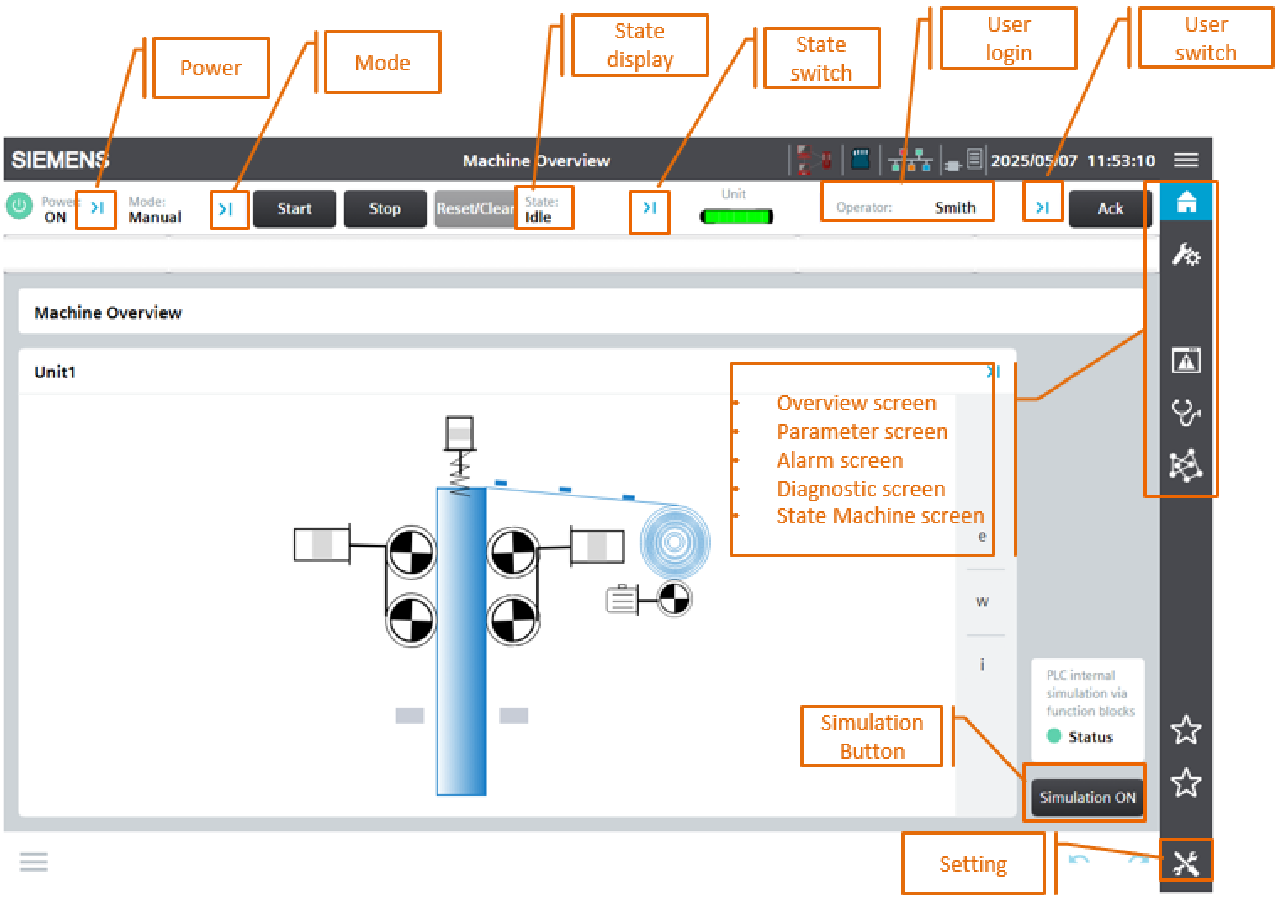


Figure 3-1: HMI Overview of the Screen

The machine overview screen Figure 3-1: HMI Overview of the Screen is the highest level of machine visualization on the HMI.

It includes:

- The Automation Framework Screen Layout, which determines how the different screens are positioned and organized in the HMI

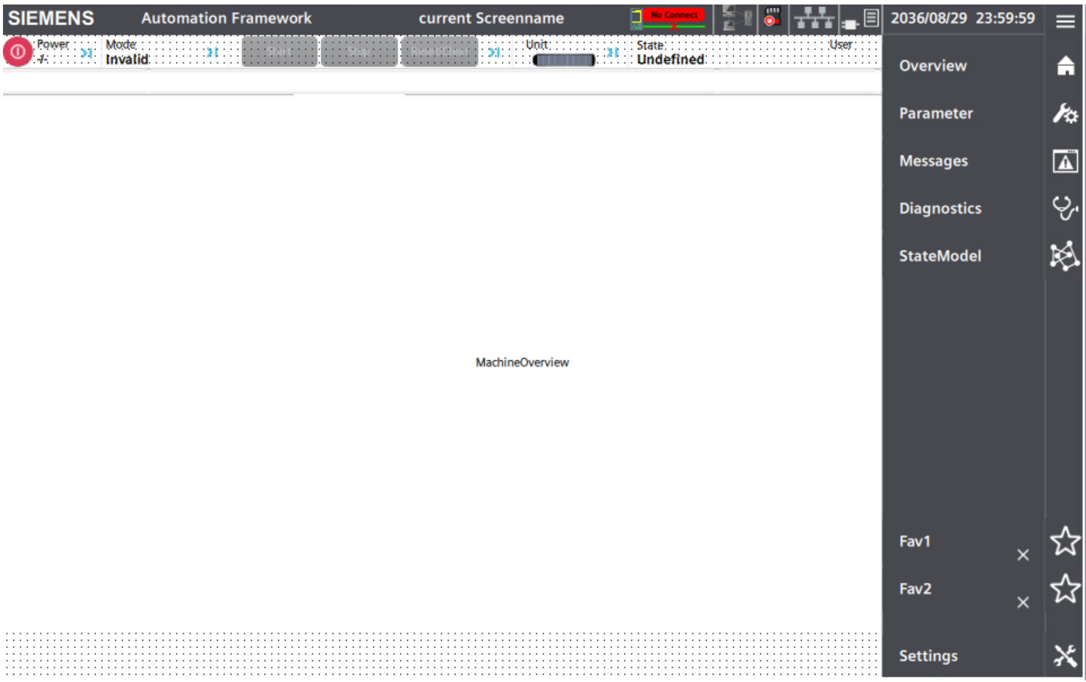


Figure 3-2: Automation Framework Screen Layout

- The header that displays the most important information (power status, current state and current mode, log in/log off dialog, Acknowledge, Date & Time, SIMATIC memory card status etc...)

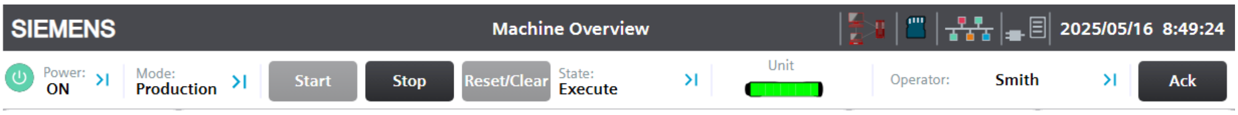


Figure 3-3: Automation Framework header

- The Main navigation that consists of two parts: The icon bar, which is permanently visible, as well as a textual view which can be switched on via the Burger menu. This navigation is pre-configured for various key features with its name and graphic accordingly. The overview (1) contains the machine overview screen. In the parameter screens (2) various use cases of configurations can be implemented (e.g., recipes, commission settings, motion information, etc.). The Message screen (3) shows current alarm/messages viewer and a historical. The Diagnostics screens (4) contains various predefined, ready-to-use diagnostic screens (system diagnostics, PROFINET diagnostics and Drives). The StateModel screen includes screens to visualize the state model and the mode state history. The Bookmarks (6) allows the user to save up to two screens to directly jump to from the main navigation and at the bottom of the list, there is the screen setting (7) for general PLC information as well as specific TIA Portal project information.

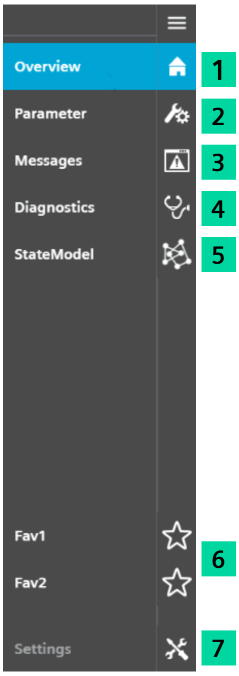


Figure 3-4: Automation Framework Main Navigation

- The Unit Navigation is created to visualize and operate the unit configured in the project.

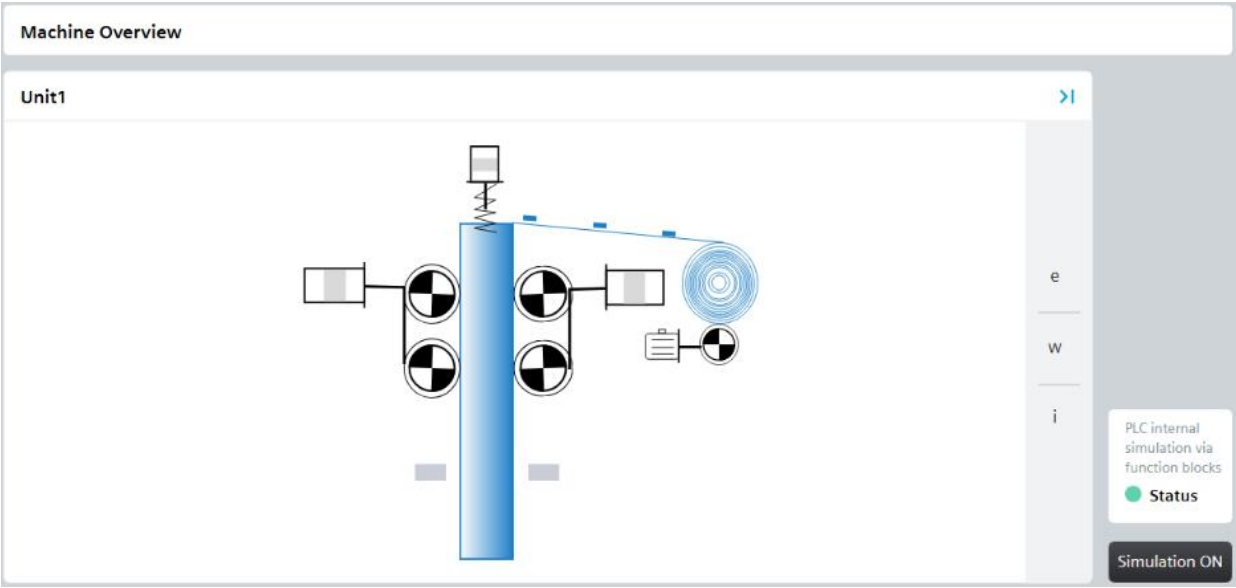


Figure 3-5: Automation Framework Unit Navigation

- The simulation button is added to the machine overview screen on the right corner. This button changes appearance based on the state.

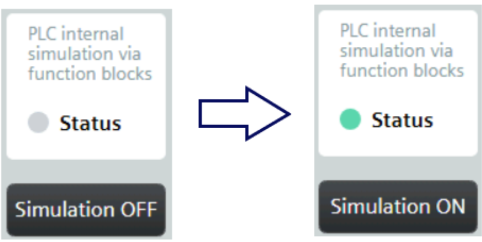


Figure 3-6: Simulation button

When the simulation button is activated, simulated signals from the color code sensor, cylinder feedback, and tension feedback are generated. These signals allow the sequence to be tested in Production mode without the need for physical hardware components.

3.2. Description of screen content

In this section we will cover how to access the different layer screens and how to make custom changes to the screens.

Machine Overview: On this screen, you can find all the information related to the Vertical Form Fill Seal Machine. When the device starts running in production mode, the main screen animation is hinted with the actual machine action.

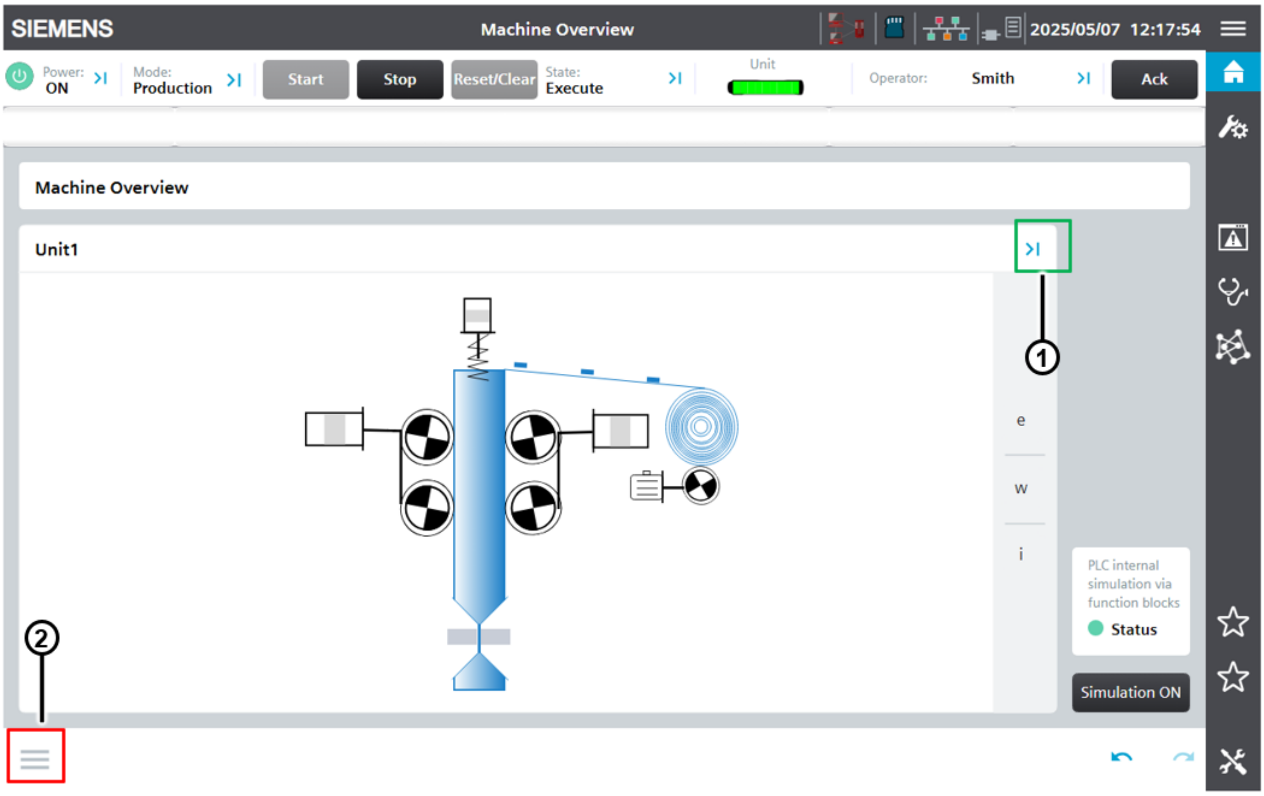


Figure 3-7: Machine overview

Unit 1 Overview: To access the overview screen of the Unit the operator can either click on the blue arrow (marked by green box in Figure 3-7: Machine overview) or on the subnavigation bar by clicking on the left button at the bottom left corner(marked by red box in Figure 3-7: Machine overview). In this screen, you can see the EMs division of the whole project and the EMs status.

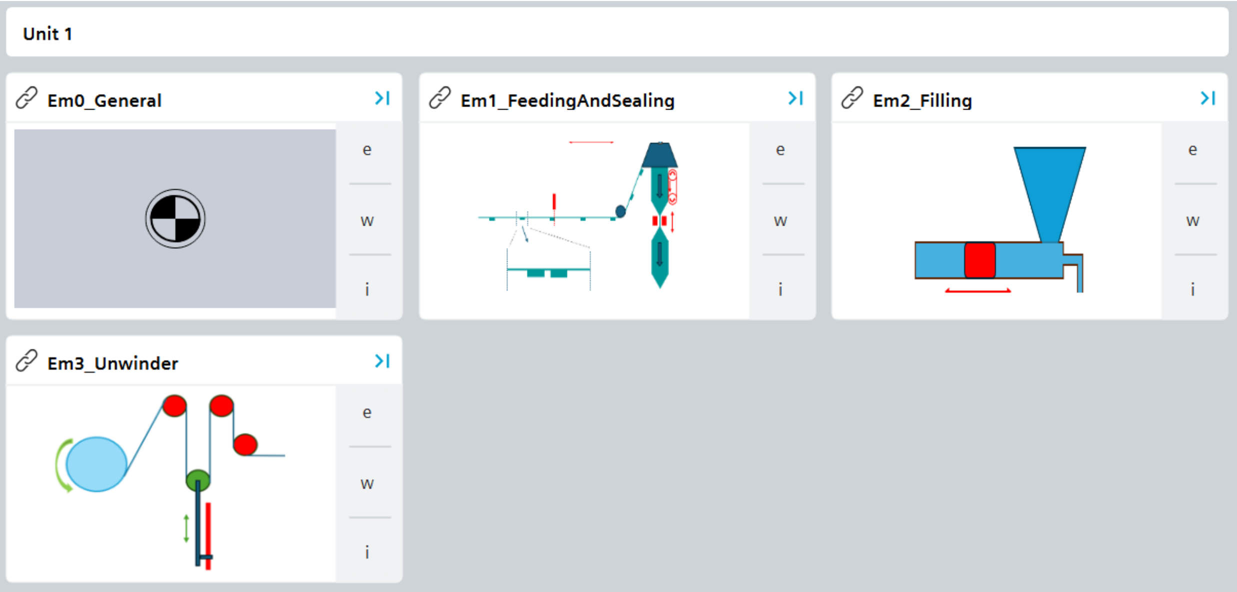


Figure 3-8: Unit 1 Overview

CM Overview: Clicking on the arrows of each EM will take you to the CM screen. This screen contains the jobs, sequence, the linking state and the CMs.

On each CM, pressing the edit button (marked by red box in Figure 3-10), a popup will be opened. Most of the popups of this machine application are faceplates from the LBC library. Parameter settings and individual operations can be performed on each CM in this popup, such as enabling, jogging, homing for axes etc. Some examples of the faceplates configured in this machine application are shown below:

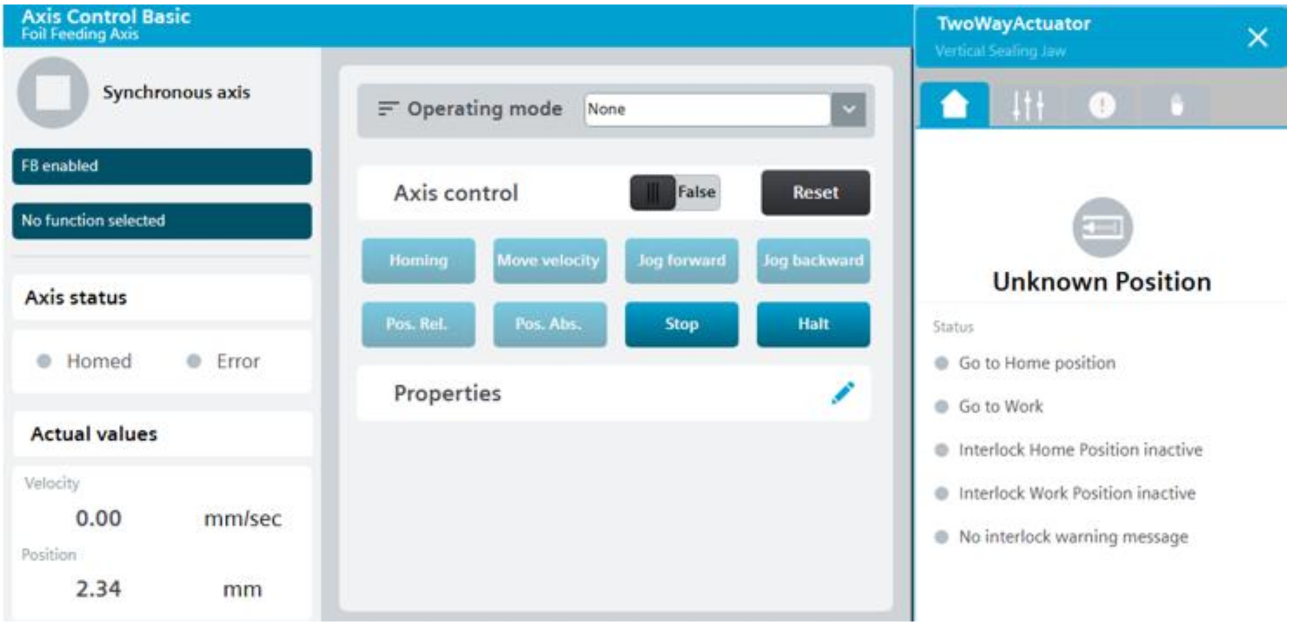


Figure 3-9: Faceplates

The faceplates for controlling the sequence, jobs and linking state can be accessed by pressing the button with the image of a hand in the bottom right corner (Figure 3-10). A popup will be opened, which is shown as Figure 3-11.

In EM0 General faceplate, you can monitor the status of the virtual master axis (e.g. velocity, position)

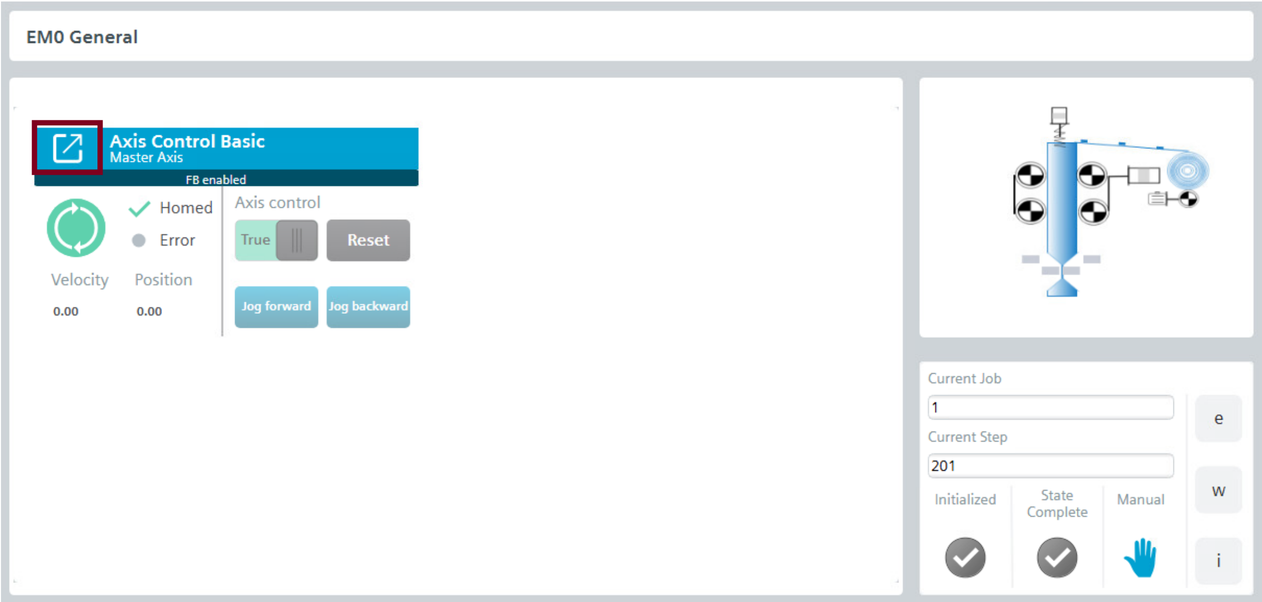


Figure 3-10: EM0 General overview

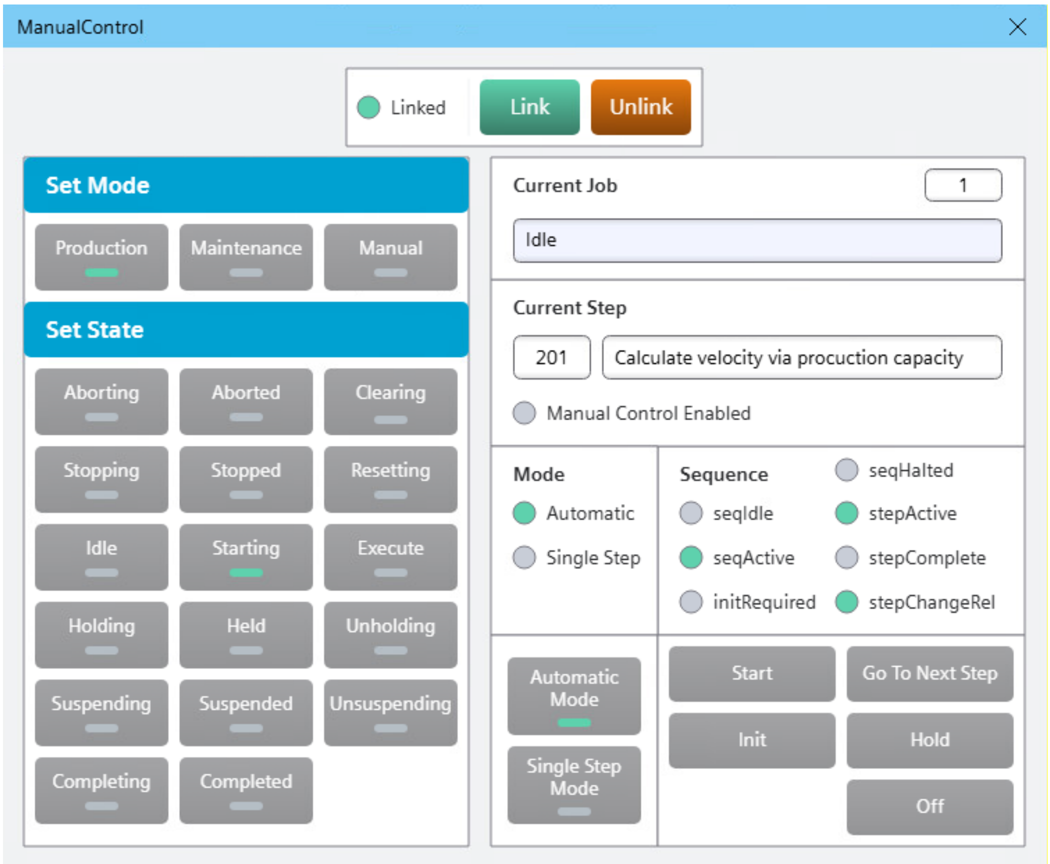


Figure 3-11: Manual Control Faceplates of the EM

In EM1 Feeding and Sealing faceplate, you can monitor and control manually the status of the Control Modules responsibles for controlling foil feeding and sealing in the machine.

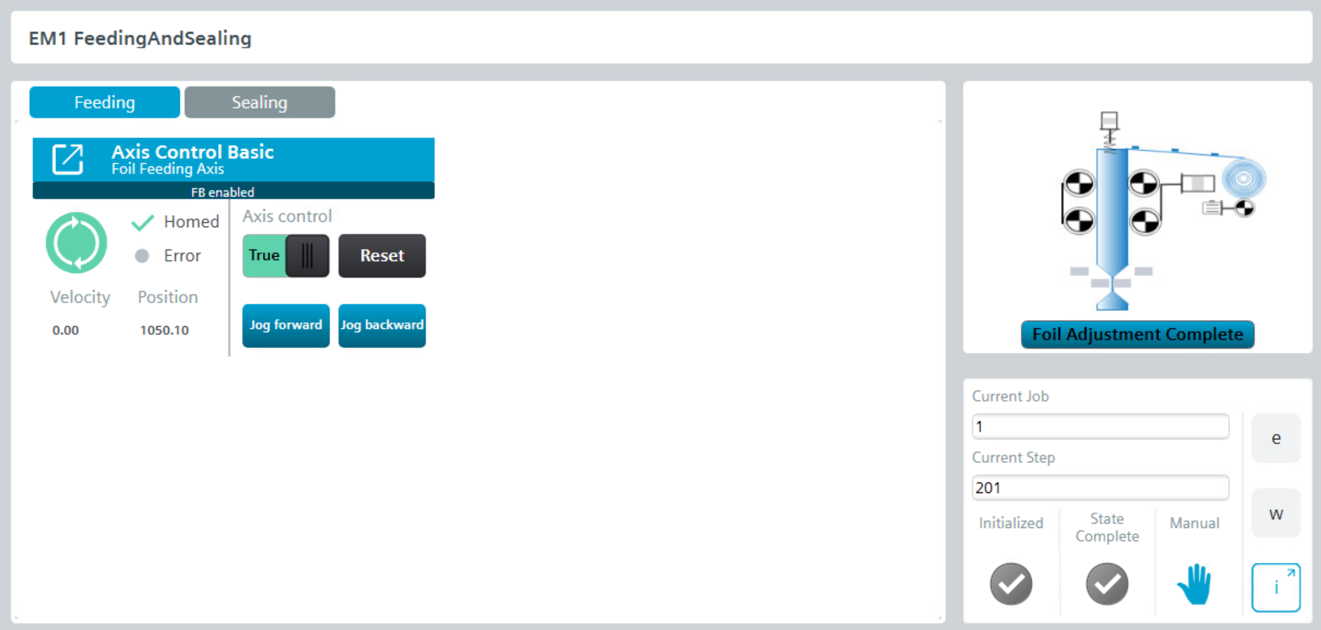


Figure 3-12: EM1 Feeding system overview

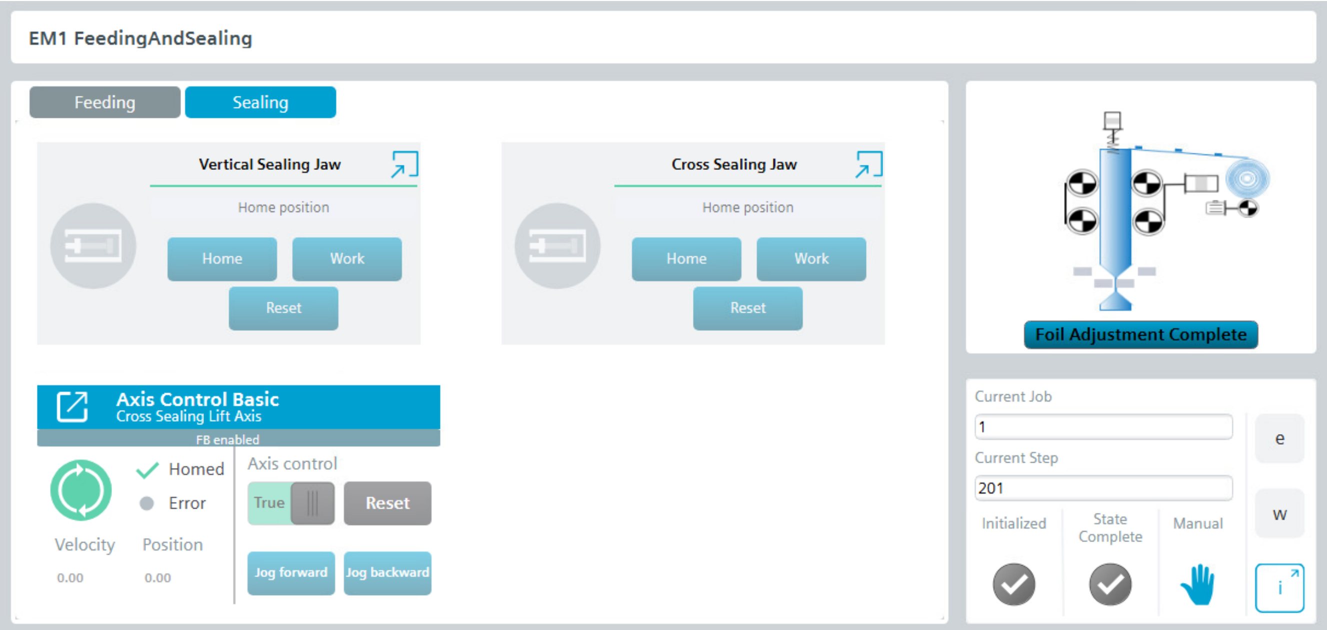


Figure 3-13: EM1 Sealing system 1

In EM2 filling faceplate, you can control manually and monitoring the status of the Control Module filling Axis Control Module, responsible for feeding the product into the machine.

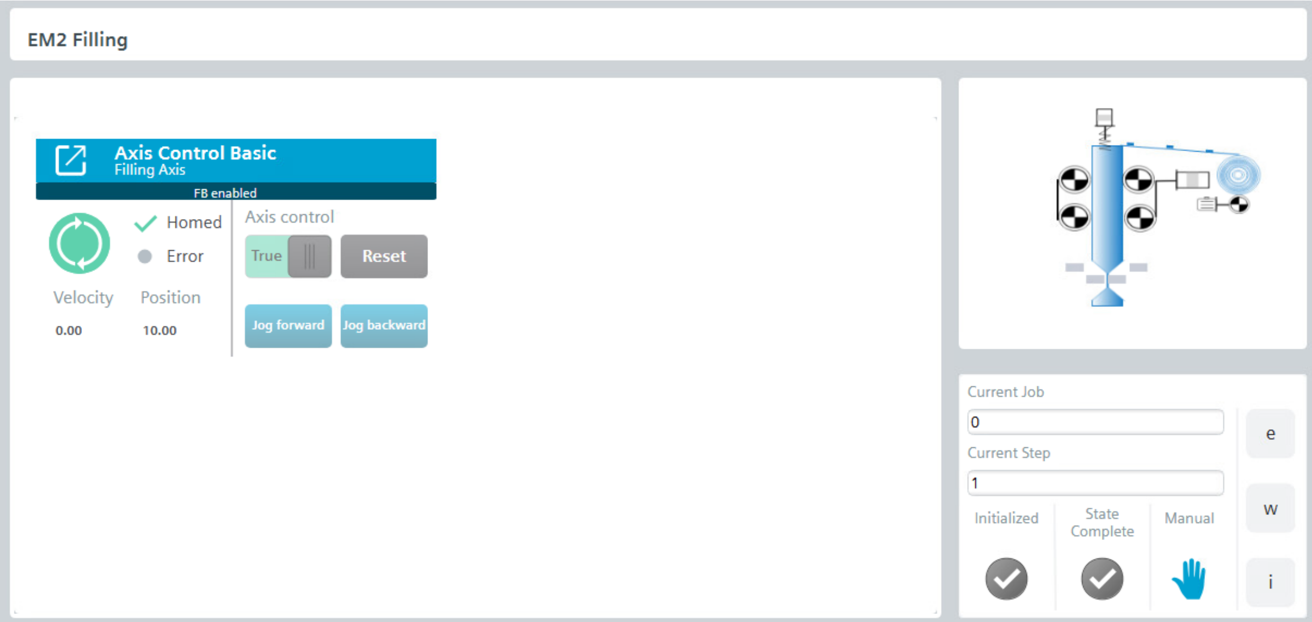


Figure 3-14: EM2 Filling system overview

In EM3 you can manually control and monitor the status of the Control Modules, which includes the unwinder motor and the analog tension sensor to ensure correct packaging.

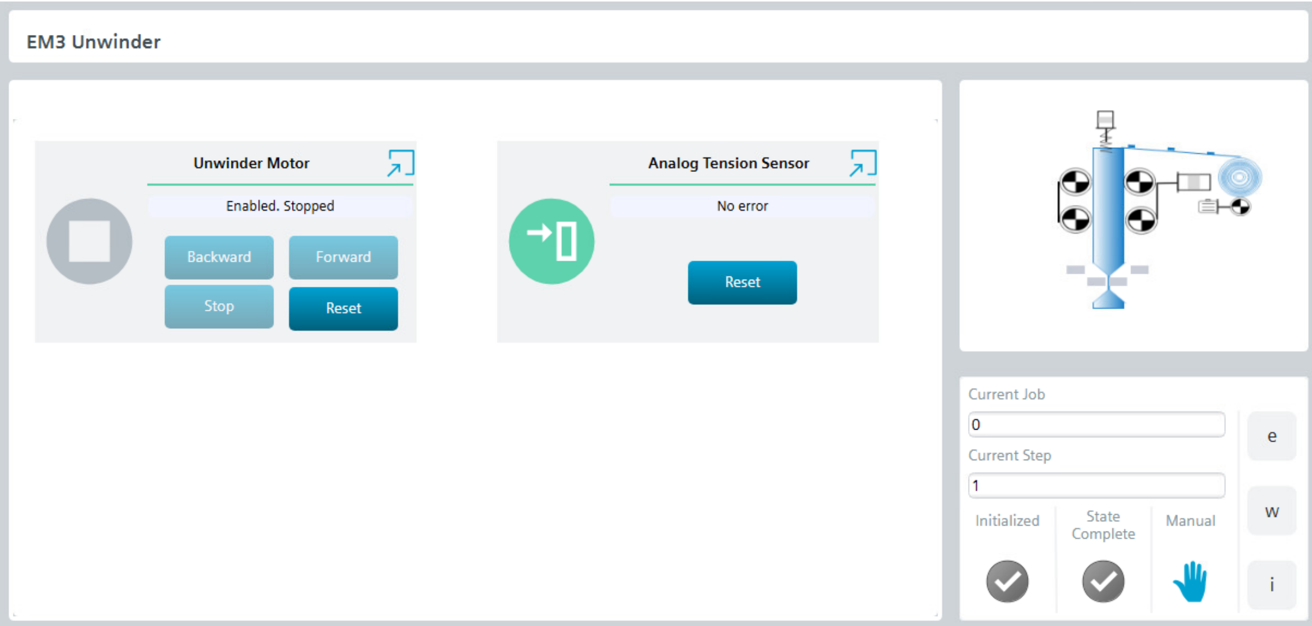


Figure 3-15: EM3 Unwinder system overview

Parameters Screen: To access this screen, click on the second icon in the main navigation, which is a wrench .

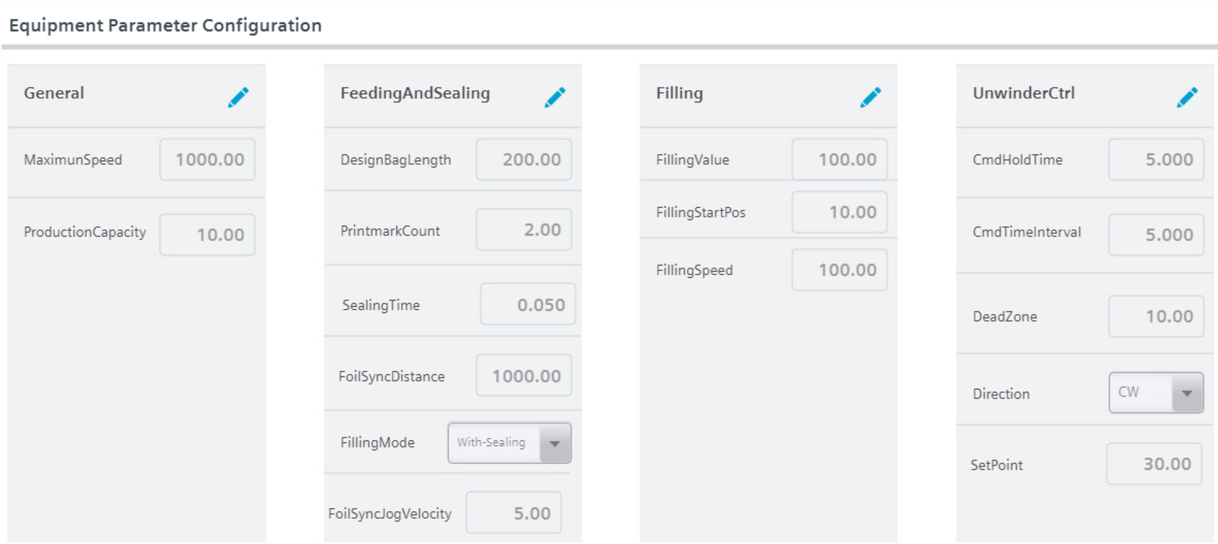


Figure 3-16: Equipment Parameter configuration overview

General	Comment
MaximumSpeed	Maximum speed for virtual axis, limited by mechanical constraints and cam track
ProductionCapacity	The expected number of bags produced per minute
DesignBagLength	The ideal bag length when designing the cam trajectory
PrintMarkCount	The number of color codes triggered by the color code sensor between each bag
SealingTime	The duration of continuous heat sealing for cross sealer jaw
FoilSyncDistance	The theoretical distance of foil feed axis that needs to be moved during the foil sync
FillingMode	Filling start with sealing or start after sealing
FoilSyncVelocity	The velocity of the foil feed axis during the foil sync process
FillingValue	The distance of the filling axis during the filling process
FillingStartPos	The position of the filling axis while waiting to fill
FillingSpeed	The velocity of the filling axis during the filling process
CmdHoldTime	The holding time for winding/unwinding output command, available when feedback is digital input
CmdTimeInterval	The interval time between winding and unwinding output command, available when feedback is digital input
DeadZone	No winding or unwinding command is output when the tension is within this range.
Direction	CW: a decrease in the analog input indicates that the tension level is reducing. CCW: an increase in the analog input indicates that the tension level is reducing. This applies only when analog feedback is used.
SetPoint	The desired tension level that indicated by the analog input signal

Table 3-1: Equipment Parameter configuration description

4. Initiating project runtime

4.1. PLC

This machine application runs on a real SIMATIC S7-1200 G2 PLC.

Download the project to the PLC

1. Download the project on your PC
2. Open it in TIA Portal V20.
3. Click on the folder icon and unzip the project into the opened folder.
4. Make sure we have a connection with the hardware PLC and select it in TIA Portal.

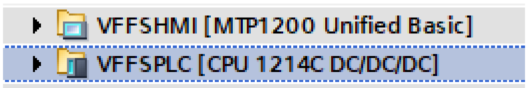


Figure 4-1: TIA Portal Screenshot

5. Click on the "Download to Device" button.
6. Select the PG/PC interface and the connection and load the project

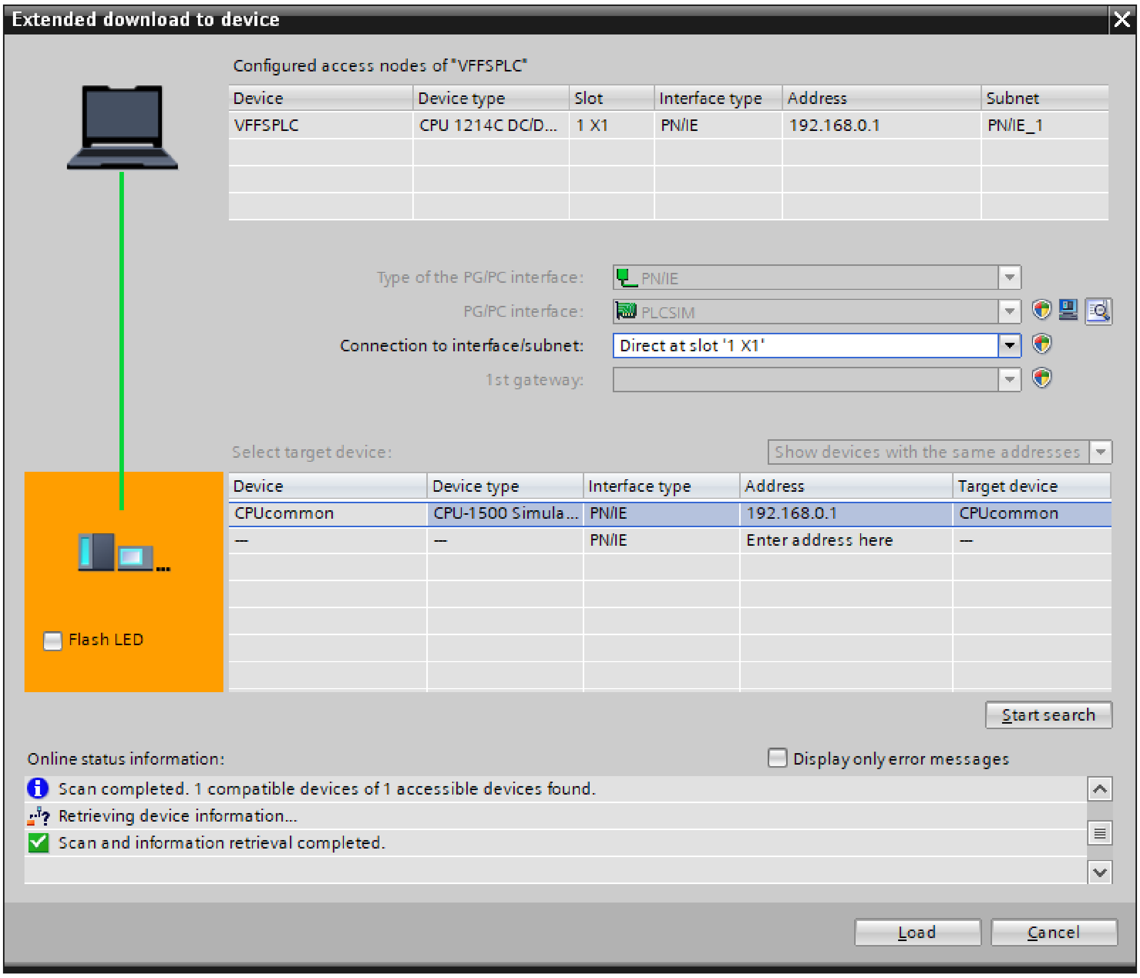


Figure 4-2: TIA Portal Screenshot

7. Load the project clicking on "Load."

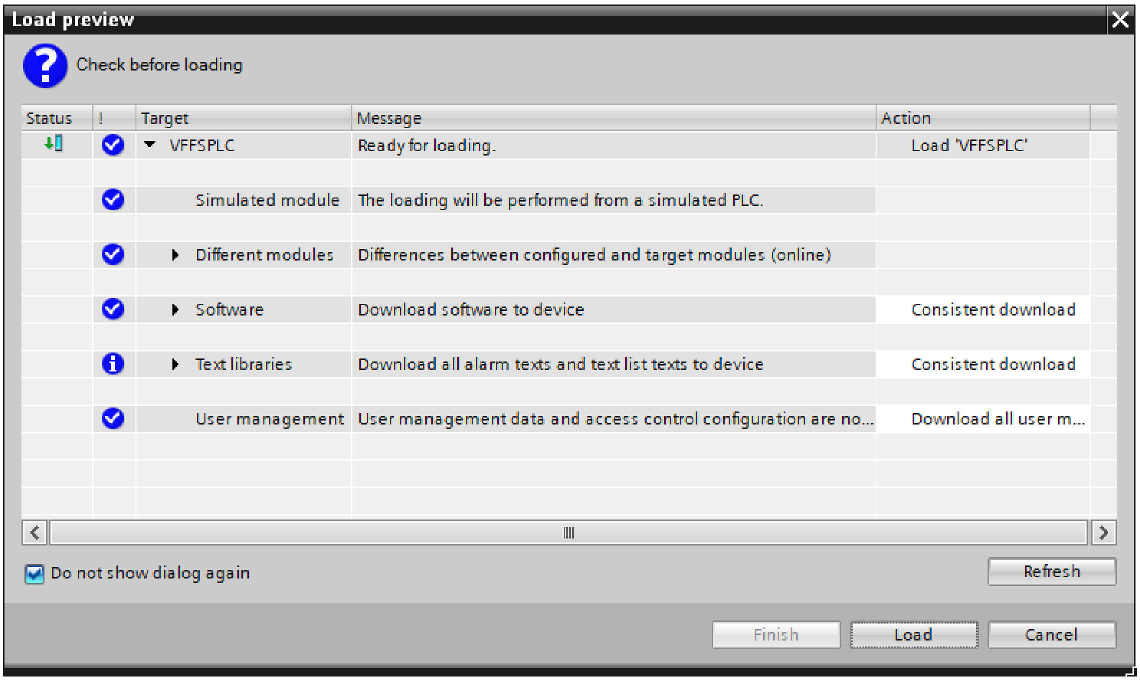


Figure 4-3:Loading Preview

8. Click on "Go Online" to ensure that the CPU is in RUN and there are no errors.

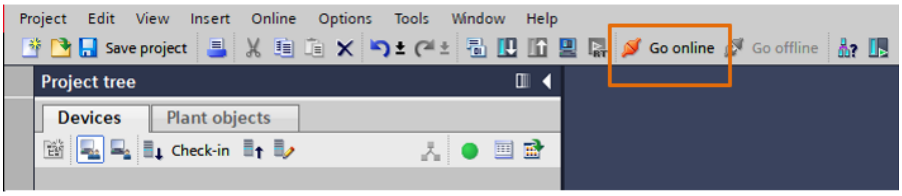


Figure 4-4: TIA Portal screenshot

4.2. HMI Runtime

Screen running can be performed by the following steps. Before running the screen, make sure the PLC program is downloaded and running without errors. Otherwise, it will not be possible to establish a connection between the HMI and the PLC.

Download the project to the HMI

1. Select the HMI VFFSHMI in TIA Portal.

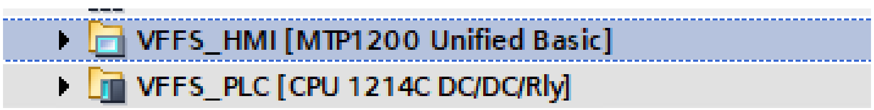


Figure 4-5: TIA Portal Screenshot

2. Download the HMI program to real HMI clicking the "Download to Device " button or clicking the "Start simulation" button to simulate the screen on the PC.

3. Press the "Request Operation" button to control the machine.

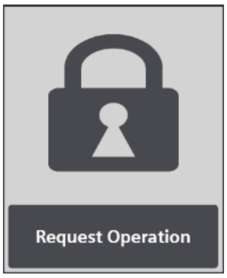
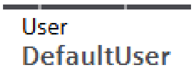


Figure 4-6: HMI screenshot

4. To log in, click on "User" 

and enter the credentials and press "Sign in":
   - Username: Jones
   - Password: Siemens!123

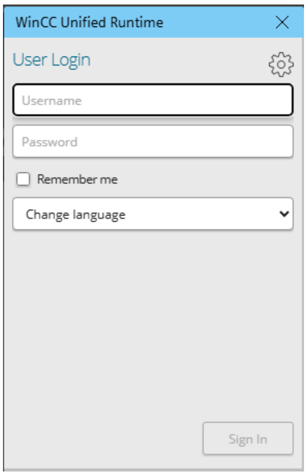


Figure 4-7: HMI screenshot

5. Click the "Mode" button to switch the mode to production mode.

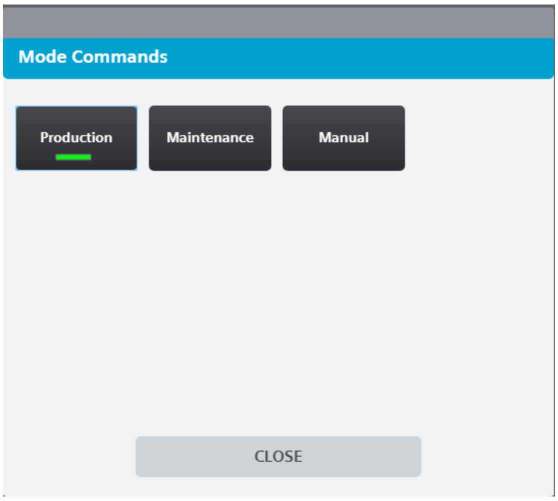


Figure 4-8:HMI screenshot

6. If you want to simulate the color code signal, cylinder signal and tension control feedback, please click the "Simulation" button. The button changes the text display to "Simulation ON", at this time the signal can be simulated by the function block

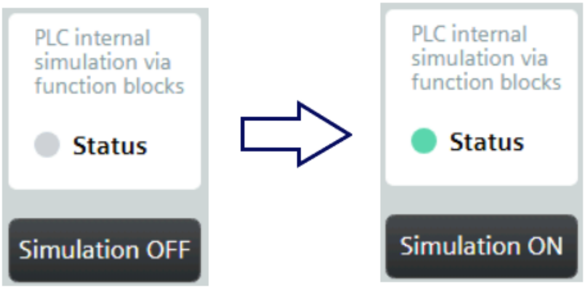


Figure 4-9: HMI screenshot

7. Click the 'Reset/Clear' button, then the 'Start' button can be clicked.

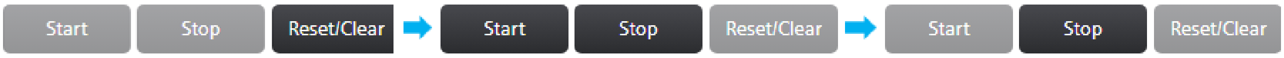


Figure 4-10: HMI screenshot

8. In the starting state, users are allowed to manually adjust the foil feeding axis to position the foil correctly. Once this adjustment is complete, the system will establish the positional relationship between the foil feeding axis and the cross sealer lift axis. Alternatively, users can skip this step, especially if high precision is not required, allowing the machine to automatically determine the positional relationship between the two axes.

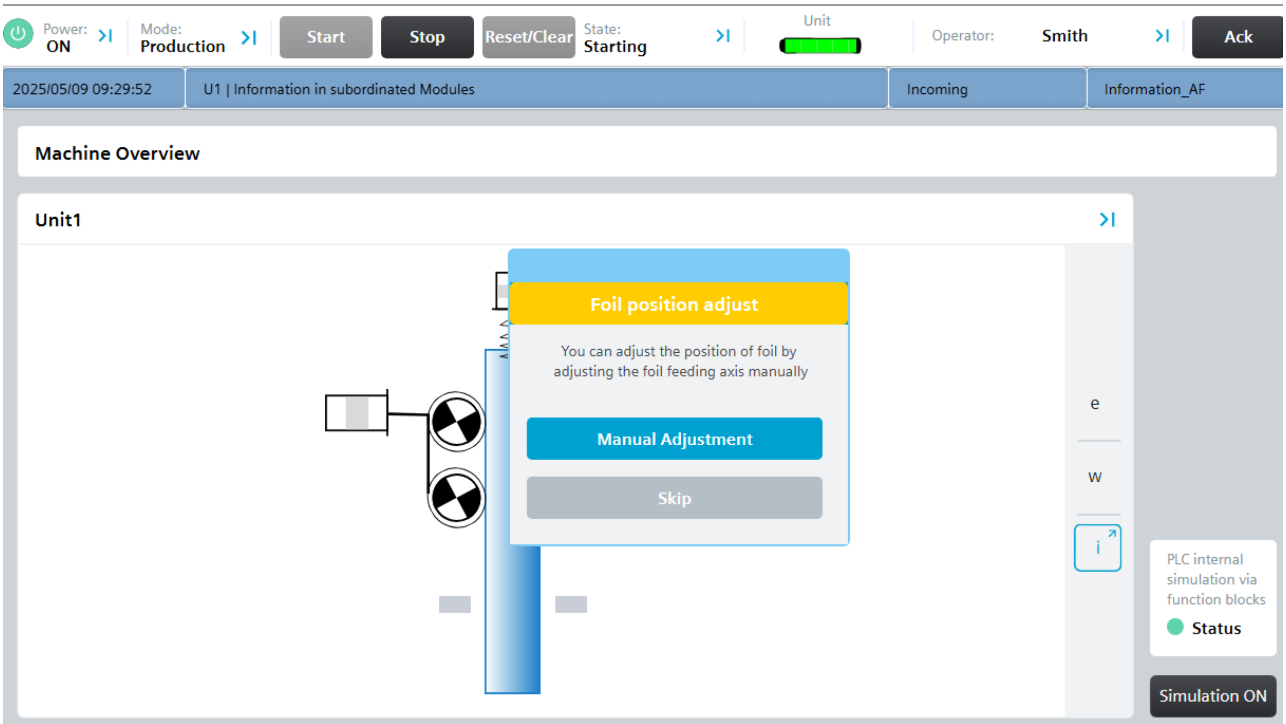


Figure 4-11:popup screen

To adjust the foil feeding axis, click the "Manual Adjustment" button. This will open the "U1Em1_FeedingAndSealing" screen, as shown below. On this screen, the user can adjust both the foil feeding axis and the cross sealing axis.

Once the adjustments are complete, click the "Foil Adjustment Complete" button to proceed.

To adjust the foil automatically, click the "Skip" button to continue the process.

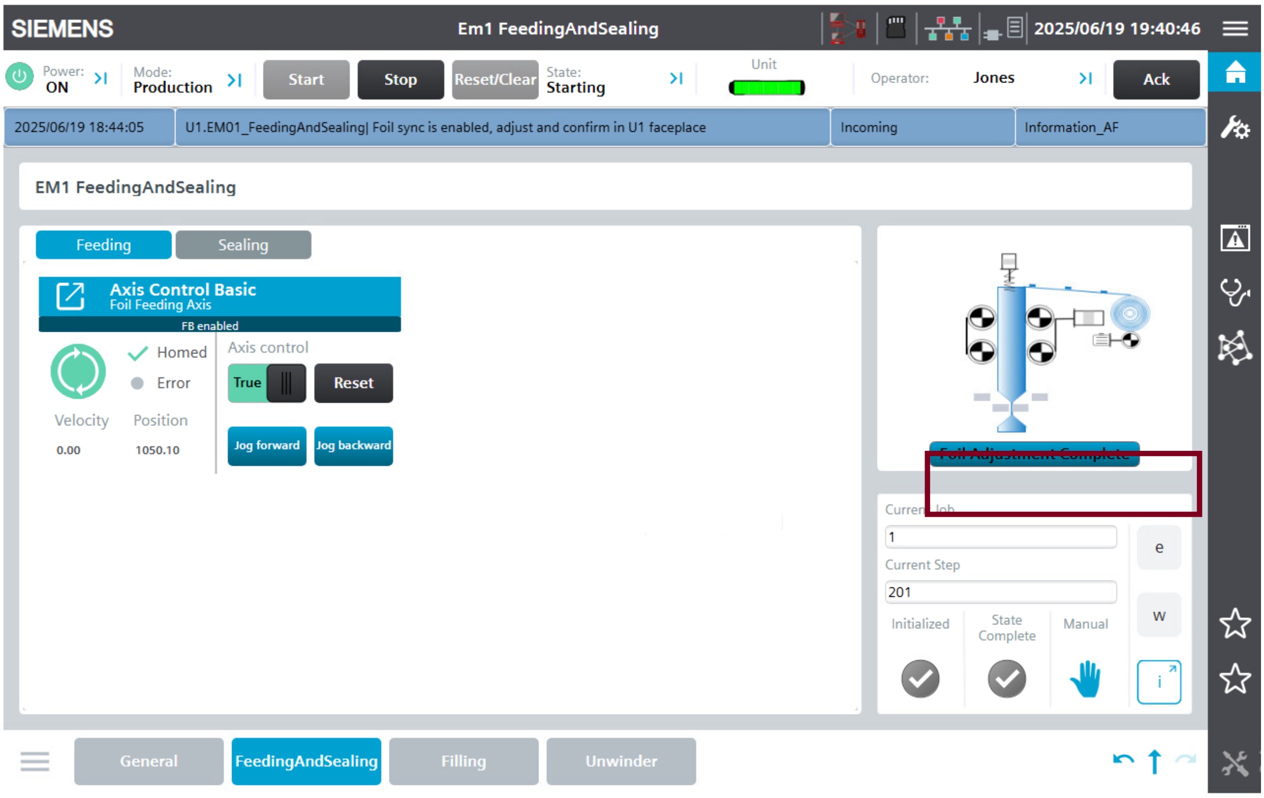


Figure 4-12:Foil Adjustment Screen

9. Once foil adjustment process is completed and the other EMs have finished the processes in the "Starting" state, the device state should displayed "Execute". Any doubts can be clarified by consulting the "State Model" screen , which shows all the necessary steps to control the machine during the entire cycle.

5. Appendix

5.1. Service and support

SiePortal

The integrated platform for product selection, purchasing and support - and connection of Industry Mall and Online support. The SiePortal home page replaces the previous home pages of the Industry Mall and the Online Support Portal (SIOS) and combines them.

- Products & Services
  In Products & Services, you can find all our offerings as previously available in Mall Catalog.
- Support
  In Support, you can find all information helpful for resolving technical issues with our products.
- mySiePortal
  mySiePortal collects all your personal data and processes, from your account to current orders, service requests and more. You can only see the full range of functions here after you have logged in.

You can access SiePortal via this address: sieportal.siemens.com

Technical Support

The Technical Support of Siemens Industry provides you fast and competent support regarding all technical queries with numerous tailor-made offers – ranging from basic support to individual support contracts.
Please send queries to Technical Support via Web form: support.industry.siemens.com/cs/my/src

SITRAIN – Digital Industry Academy

We support you with our globally available training courses for industry with practical experience, innovative learning methods and a concept that's tailored to the customer's specific needs.
For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page: siemens.com/sitrain

Industry Online Support app

You will receive optimum support wherever you are with the "Industry Online Support" app. The app is available for iOS and Android:

Available on the App Store

ANDROID APP ON Google play

5.2. Links and literature

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to this entry page of this application example https://support.industry.siemens.com/cs/ww/en/view/109989389
\3\	Programming Style Guide https://support.industry.siemens.com/cs/ww/en/view/81318674
\4\	Standardized guidelines https://support.industry.siemens.com/cs/ww/en/view/109756737
\5\	Guidelines for library processing in the TIA Portal https://support.industry.siemens.com/cs/ww/en/view/109747503
\6\	Automation Framework https://support.industry.siemens.com/cs/ww/en/view/109817223

5.3. Change documentation

Version	Date	Modification
V1.0	07/2025	First version

